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VALIDATION RESULTS
OF PHU MY GDC GAS METERING SYSTEM
PHU MY 4 POWER PLANT

Contract No.:

223/2019/DVK/TMHĐ-IAS

Ngày 06 tháng 06 năm 2019

2020

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UNCERTAINTY CERTIFICATES

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GIẤY CHỨNG NHẬN ĐỘ KHÔNG ĐẢM BẢO ĐO

CERTIFICATE OF UNCERTAINTY

Số/ *No.* : **KT3-01325CDT9/1**

1. Phương tiện đo/ *Instrument* : **Hệ thống đo khí thiên nhiên/ *Natural gas metering system***
2. Kiểu/ *Type* : **Đồng hồ sóng siêu âm/ *Ultrasonic Flow Meter***
3. Mã số thiết bị/ *Tag name* : FT-1420
4. Đặc trưng kỹ thuật/ *Technical specifications* :

   Đường kính đồng hồ/ *Inner diameter of meter* : 10 inches

   Dải lưu lượng/ *Range* : đến/*up to* 5 000 m^3/h

5. Danh sách thiết bị trên hệ thống/ *List of equipments* :

Equipment	Manufacture	ID.	Series no.	Calib. date	Due date
Máy tính lưu lượng *Flow computer*	Daniel/ FloBoss S600+	FQI-1420	-	20/12/2019	N/A
Đồng hồ siêu âm *Ultrasonic flow meter*	Daniel	FT-1420	02-050175	18/02/2020	N/A
Bộ chuyển đổi đo áp suất *Pressure transmitter*	Rosemount	PT-1423	RS 0911811	05/03/2019	31/03/2020
Bộ chuyển đổi đo nhiệt độ *Temperature transmitter*	Rosemount	TT-1424	12025577	06/03/2019	31/03/2020
Máy sắc ký khí *Gas analysis*	Daniel	AIC-1421	UK22R1087	19/12/2019	N/A

6. Khách hàng/ *Customer* : **Công ty Vận chuyển khí Đông Nam Bộ - PV Gas SE**
   Số 101 Lê Lợi, P. Thắng Nhì, TP. Vũng Tàu, Tỉnh BR-VT
7. Nơi sử dụng/ *Place of working* : **Trung tâm phân phối khí Phú Mỹ - Phu My GDC**
   Khu CN Phú Mỹ 1, thị trấn Phú Mỹ, huyện Tân Thành, Tỉnh BR - VT
8. Phương pháp đánh giá / *Method of validation* : QTHC/KT3 08 - Hệ thống đo khí bằng đồng hồ sóng siêu âm và turbine (AGA9; AGA8; ISO 6976; ISO 5168)
9. Ngày thực hiện/ *Validated date* : **12/2019 - 03/2020**
10. Ngày cấp/ *Issued date* : **11/03/2020**

PHỤ TRÁCH PĐL DUNG TÍCH LƯU LƯỢNG
Head of Volume & Flow Measurement Lab.

Nguyễn Văn Duy

KT. GIÁM ĐỐC, PHÓ GIÁM ĐỐC
For Director/ Vice Director

Trương Thanh Sơn

11. Độ không đảm bảo đo của hệ thống/ *Uncertainty of metering system* :

Lưu lượng/ *Flow rate*				Độ không đảm bảo đo/ *Uncertainty*	
% Q_{max}	Danh định/ *Nominal*	Thể tích chuẩn/ *Standard volume*	Năng lượng/ *Energy*	Thể tích chuẩn/ *Standard volume*	Năng lượng/ *Energy*
%	m^3/h	Sm^3/h	MJ/h	%	%
5,0	250	8818	358578	0,35	0,36
10,1	504	17739	721339	0,36	0,37
25,2	1262	44473	1808490	0,36	0,37
39,9	1996	70330	2859928	0,35	0,36
69,8	3488	122884	4997013	0,34	0,36
80,4	4022	141678	5761257	0,35	0,36

Note :

- Độ không đảm bảo đo được tính theo ISO 5168:2005 với phân bố chuẩn tương ứng độ tin cậy 95% và hệ số phủ k =2/ The total uncertainty is calculated base on ISO 5168:2005 with the confident level at 95 % and coverage factor k = 2;

- Kết quả tính toán dựa trên điều kiện làm việc của hệ thống tại áp suất chân không 34,01325 bar a và 20° C/ The calculated results in the operating condition at absolute pressure 34,01325 bar a and 20° C;

- Thành phần khí áp dụng tính toán tương ứng với thành phần khí làm việc của hệ thống/ The applied gas composition was representative of working system natural gas.

- Đơn vị lưu lượng khí chuẩn được tính tại điều kiện chuẩn là áp suất chân không 1,01325 bar a và 15° C/ The unit of volume flowrate is converted to the reference condition at 1,01325 bar a and 15° C

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GIẤY CHỨNG NHẬN ĐỘ KHÔNG ĐẢM BẢO ĐO

CERTIFICATE OF UNCERTAINTY

Số/ *No.* : **KT3-01325CDT9/2**

1. Phương tiện đo/ *Instrument* : **Hệ thống đo khí thiên nhiên/ *Natural gas metering system***
2. Kiểu/ *Type* : **Đồng hồ sóng siêu âm/ *Ultrasonic Flow Meter***
3. Mã số thiết bị/ *Tag name* : FT-1440
4. Đặc trưng kỹ thuật/ *Technical specifications* :

   Đường kính đồng hồ/ *Inner diameter of meter* : 10 inches

   Dải lưu lượng/ *Range* : đến/*up to* 5 000 m^3/h

5. Danh sách thiết bị trên hệ thống/ *List of equipments* :

Equipment	Manufacture	ID.	Series no.	Calib. date	Due date
Máy tính lưu lượng *Flow computer*	Daniel/ FloBoss S600+	FQI-1440	-	20/12/2019	N/A
Đồng hồ siêu âm *Ultrasonic flow meter*	Daniel	FT-1440	02-050174	26/11/2019	N/A
Bộ chuyển đổi đo áp suất *Pressure transmitter*	Rosemount	PT-1443	2732618	05/03/2019	31/03/2020
Bộ chuyển đổi đo nhiệt độ *Temperature transmitter*	Rosemount	TT-1444	1067571	05/03/2019	31/03/2020
Máy sắc ký khí *Gas analysis*	Daniel	AIC-1421	UK22R1087	19/12/2019	N/A

6. Khách hàng/ *Customer* : **Công ty Vận chuyển khí Đông Nam Bộ - PV Gas SE**
   Số 101 Lê Lợi, P. Thắng Nhì, TP. Vũng Tàu, Tỉnh BR-VT
7. Nơi sử dụng/ *Place of working* : **Trung tâm phân phối khí Phú Mỹ - Phu My GDC**
   Khu CN Phú Mỹ 1, thị trấn Phú Mỹ, huyện Tân Thành, Tỉnh BR - VT
8. Phương pháp đánh giá / *Method of validation* : QTHC/KT3 08 - Hệ thống đo khí bằng đồng hồ sóng siêu âm và turbine (AGA9; AGA8; ISO 6976; ISO 5168)
9. Ngày thực hiện/ *Validated date* : **12/2019 - 03/2020**
10. Ngày cấp/ *Issued date* : **11/03/2020**

PHỤ TRÁCH PĐL DUNG TÍCH LƯU LƯỢNG	**KT. GIÁM ĐỐC, PHÓ GIÁM ĐỐC**
Head of Volume & Flow Measurement Lab.	*For Director/ Vice Director*
Nguyễn Văn Duy	**Trương Thanh Sơn**

11. Độ không đảm bảo đo của hệ thống/ *Uncertainty of metering system* :

Lưu lượng/ *Flow rate*				**Độ không đảm bảo đo/ *Uncertainty***	
% Q_{max}	**Danh định/ *Nominal***	**Thể tích chuẩn/ *Standard volume***	**Năng lượng/ *Energy***	**Thể tích chuẩn/ *Standard volume***	**Năng lượng/ *Energy***
%	**m^3/h**	**Sm^3/h**	**MJ/h**	**%**	**%**
5,0	**248**	**8729**	**354968**	**0,36**	**0,38**
10,0	**500**	**17624**	**716655**	**0,36**	**0,37**
25,0	**1248**	**43957**	**1787475**	**0,37**	**0,39**
40,3	**2016**	**71019**	**2887963**	**0,38**	**0,39**
69,8	**3492**	**123026**	**5002801**	**0,37**	**0,38**
79,8	**3988**	**140487**	**5712851**	**0,37**	**0,39**

Note :

- Độ không đảm bảo đo được tính theo ISO 5168:2005 với phân bố chuẩn tương ứng độ tin cậy 95% và hệ số phủ k =2/ The total uncertainty is calculated base on ISO 5168:2005 with the confident level at 95 % and coverage factor k = 2;

- Kết quả tính toán dựa trên điều kiện làm việc của hệ thống tại áp suất chân không 34,01325 bar a và 20 ^{o} C/ The calculated results in the operating condition at absolute pressure 34,01325 bar a and 20 ^{o} C;

- Thành phần khí áp dụng tính toán tương ứng với thành phần khí làm việc của hệ thống/ The applied gas composition was representative of working system natural gas.

- Đơn vị lưu lượng khí chuẩn được tính tại điều kiện chuẩn là áp suất chân không 1,01325 bar a và 15 ^{o} C/ The unit of volume flowrate is converted to the reference condition at 1,01325 bar a and 15 ^{o} C

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UNCERTAINTY ANALYSIS

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UNCERTAINTY ANALYSIS OF GAS METERING SYSTEM IN PHU MY GDC PHU MY 4 POWER PLANT

CONTRACT No.: KT3-01325CDT9

Prepared by	Checked by	Approval by
Nguyen Nhu Y *Metering Engineer*	**Nguyen Van Duy** *Head of Volume & Flow Measurement Lab.*	**Truong Thanh Son** *Vice Director*

Revision : 1

REVISION CONTROL

Review	Date of issue	Description	Prepared	Checked	Approved by
0		Document Development	Nguyen Nhu Y	Nguyen Van Duy	
1	11/03/2020	Issued	Nguyen Nhu Y	Nguyen Van Duy	Truong Thanh Son

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3. ULTRASONIC METER UNCERTAINTY

4. STANDARD VOLUME FLOWRATE UNCERTAINTY

5. ENERGY FLOWRATE UNCERTAINTY

6. SUMMARY OF RESULTS

7. REFERENCES

1.0 INTRODUCTION

The purpose of this estimation study is to establish the uncertainty of of Phu My Gas Distribution Centre (GDC). The uncertainty estimation has been performed at actual line condition then transferred to the base condition at 15 ^{o}C and 1,01325 bar a. The uncertainty of the system will be presented below.

2.0 SYSTEM DESCRIPTION

The custody transfer gas metering skid of Phu My GDC is Ultrasonic (USM) metering system.

Each system consists of two Ultrasonic flow meters, each sized for 100% flow 5 000 m^3/h, connected in series with bypass lines around them on a skid. In normal use, one of them is used as the operation meter and the other as standby meter. Normally the flow shall be through the operation meter and shall bypass the standby meter, by manually closing the bypass line. During operation meter maintenance, the standby meter shall be used as the operation meter.

Each meter run is equipped with one Daniel FloBoss S600 flow computer for flow computation with built in pressure transmitter. Temperature transmitter is used for line gas temperature measurement with a (4-20) mA analogue input into flow computer. The Daniel 500 Series gas chromatograph (GC) is installed at the Phu My GDC main control terminal for line gas composition analyses. GC data is distributed into each flow computer via MODBUS protocol through RS-485 serial link. Flow computer shall use the received gas data and normalized it for energy calculation.

Line volume flowrate is calculated in accordance with AGA 9 for ultrasonic meter. Line compressibility is calculated in accordance with AGA 8. Standard volume flow is then derived with the PTZ formula and energy flow rate is obtained with the calorific value calculated using ISO 6976.

3.1. Flow calculation methodology

The flow calculation methodologies for Phu My GDC - Gas metering system are in accordance with AGA 9 for ultrasonic meter, AGA 8-1994, and ISO 6976-1995.

3.2. Ultrasonic metering system

As refer to AGA 9, the flow process data measured by ultrasonic meter can be transmitted to flow computer via MODBUS protocol through RS 485 serial link. In this project, a digital signal is selected. The line volume flow rate is then given as below

$$Q_{line} = \frac{f}{KF_{meter}} \times 3600 \times C_{Tsm} \times C_{Psm} \qquad (1)$$

and the standard volume flowrate and energy flowrate is illustrated as following equations (in accordance with AGA 9 Section 7.3.1)

$$Q_{std} = Q_{line} \times \frac{\rho_l}{\rho_{std}} \qquad (2)$$

$$Q_m = \frac{1}{24} \times \frac{Q_{line} \times \rho_l}{KF} \qquad (3)$$

$$Q_E = Q_{std} \times Hs_{vol} \qquad (4)$$

$$VCF = \left(\frac{P_l}{P_b}\right) \times \left(\frac{T_b}{T_l}\right) \times \left(\frac{Z_b}{Z_l}\right) = \left(\frac{\rho_l}{\rho_b}\right) \qquad (5)$$

where

Q_{line} Volume flow rate at line condition, m^3/h

Q_{std} Volume flow rate at base condition, m^3/h

Q_m Mass flow rate, kg/h

Q_E Energy flow rate, MJ/h

KF_{meter} Meter K-Factor derived from calibration certificate

f Frequency/pulses generated from ultrasonic meter

Hs_{vol} Volume base heating value calculated in accordance with ISO 6976 - 1995, MJ/m^3

VCF Volume correction factor to base condition.

P_l Pressure at line condition, bar a

P_b Pressure at base condition, 1,01325 bar a

T_l Temperature at line condition, °C

T_b Temperature at base condition, 15 °C (288,15 K)

Z_l Gas compressibility at line condition, calculated in accordance with AGA 8-1994

Z_b Gas compressibility at base condition, calculated in accordance with ISO 6976-1995

ρ_l Gas density at line condition, calculated in accordance with AGA 8 - 1994

ρ_b Gas density at base condition, calculated in accordance with ISO 6976-1995

3.3 Derivation of ultrasonic metering uncertainty

The ultrasonic meter was sent for a wet calibration at a third party calibration facility in CEESI. From the calibration certificate, CEESI had declared that the maximum tolerance error of the calibrated meter is 0,004 % for FT 1420, 0,05 % for FT 1440.

From the experimental result of the calibration, the maximum calibration uncertainty (U_{cal}) of meters which will affect the actual measurement during operation is estimated for this experimental data 0,18 % of calibrated span.

An installation configuration error will contributed to the meter installation at site (U_{mi}). AGA 9 Section 7.2.2. recommended that a maximum error is 0,3 % for this effect and it has accounted for the k multiplier.

The USM combined uncertainty is given a follow

$$U(m) = \sqrt{\left(\frac{U_{cal}}{2}\right)^2 + \left(\frac{U_{tol}}{1,73}\right)^2 + \left(\frac{U_{mi}}{2,45}\right)^2}$$

The calculated expanded uncertainty for ultrasonic meter is summarized in tables below

- **For FT – 1420**

Ultrasonic meter flow range (%)	Standard uncertainty (%)			Combined uncertainty	Expanded uncertainty k = 2
	U_{cal}	U_{tol}	U_{mi}		
5,0	**0,15**	**0,000**	**0,30**	0,14	0,29
10,1	**0,17**	**0,000**	**0,30**	0,15	0,30
25,2	**0,17**	**0,000**	**0,30**	0,15	0,30
39,9	**0,15**	**0,000**	**0,30**	0,14	0,29
69,8	**0,14**	**0,000**	**0,30**	0,14	0,28
80,4	**0,15**	**0,000**	**0,30**	0,14	0,29
ci	1,00	1,00	1,00		

- **For FT – 1440**

Ultrasonic meter flow range (%)	Standard uncertainty (%)			Combined uncertainty	Expanded uncertainty k = 2
	U_{cal}	U_{tol}	U_{mi}		
5,0	0,14	0,000	0,30	0,14	0,28
10,0	0,13	0,000	0,30	0,14	0,28
25,0	0,17	0,000	0,30	0,15	0,30
40,3	0,18	0,000	0,30	0,15	0,30
69,8	0,15	0,000	0,30	0,14	0,29
79,8	0,17	0,000	0,30	0,15	0,30
ci	1,00	1,00	1,00		

4.0 STANDARD VOLUME FLOWRATE UNCERTAINTY

The standard flowrate is illustrated as following equation(in accordance with AGA 9 Section 7.3.1)

$$Q_{std} = Q_{line} \times VCF$$

$$VCF = \left(\frac{P_l}{P_b}\right) \times \left(\frac{T_b}{T_l}\right) \times \left(\frac{Z_b}{Z_l}\right) = \left(\frac{\rho_l}{\rho_b}\right)$$

the uncertainty equation of standard volume flow for both elements is therefore,

$$U_{qstd} = \sqrt{\left(\frac{U_m}{2}\right)^2 + \left(\frac{U_{(Rho)\exp}}{2}\right)^2 + \left(\frac{U_{(Rhostd)}}{2}\right)^2 + \left(\frac{U_{(fc)\exp}}{2}\right)^2}$$

4.1 Upstream density

The upstream density is calculated from the gas composition measured by a gas chromatograph Daniel 500 series, the line pressure measured by a pressure transducer and the line temperature measured by a temperature RTD,

The functional relationship used to determine the line density is defined in AGA8 – 1994, this can be written in the form:

$$y = f(x_1, x_2, x_3 \ldots x_i)$$

where the output y (density) is a function of a number of inputs x (components, temperature and pressure values)

The sensitivity coefficients for each of the input quantities are calculated by successively increasing and decreasing each input by the magnitude of its estimated uncertainty and calculating the effect this has on the resulting density as described in the GUM[1]

A temporary variable Z is first calculated

$$Z_i = \frac{1}{2}\left[f\left(x_1, \ldots x_i + u(x_i), \ldots, x_n\right) - f\left(\left(x_1, \ldots, x_i - u(x_i), \ldots, x_n\right)\right)\right]$$

and where the expanded uncertainty is expressed in the form:

$$u_c(y) = \sqrt{\sum_{i=1}^{N} c_i^2 \cdot u^2(x_i)}$$

the value of each sensitivity coefficient c_i taken as $Z_i/u(x_i)$.

By using the line pressure, temperature and live gas data from gas chromatograph, the line compressibility factor and also the line density value can be obtained in accordance to AGA8 - 94 calculation. The combined standard uncertainty is calculated as show below

[1] *Guide to the Expression of Uncertainty in Measurement*, Section 5

$$u_{(Rho)} = \sqrt{\left(\frac{U_{(p)\exp}}{2}\right)^2 + \left(\frac{U_{(t)\exp}}{2}\right)^2 + \left(\frac{U_Z}{2}\right)^2 + \left(\frac{U_{N_2}}{2}\right)^2 + \left(\frac{U_{CO_2}}{2}\right)^2 + \left(\frac{U_{C1}}{2}\right)^2 + \left(\frac{U_{C2}}{2}\right)^2 + \left(\frac{U_{C3}}{2}\right)^2 + \left(\frac{U_{i-C4}}{2}\right)^2 + \left(\frac{U_{n-C4}}{2}\right)^2 + \left(\frac{U_{i-C5}}{2}\right)^2 + \left(\frac{U_{n-C5}}{2}\right)^2 + \left(\frac{U_{C6+}}{2}\right)^2}$$

The expanded uncertainty is

$$U_{(Rho)\exp} = 2 \times u_{Rho}$$

4.1.1 Pressure

Line pressure is measured using a type pressure transmitter built in PT 1423, PT 1443. The pressure transducer was calibrated on field for a span (0 – 60,0) barg . Line pressure was estimated at operating pressure of 34,01325 bar a

The combined standard uncertainty in pressure is estimated by combining inputs derived from the calibration of the transducer and the manufacturer's specification using the following equation:

$$u_p = \sqrt{\left(\frac{U_{pac}}{\sqrt{3}}\right)^2 + \left(\frac{U_{pcal}}{2}\right)^2 + \left(\frac{U_{ptemp}}{\sqrt{3}}\right)^2 + \left(\frac{U_{padc}}{\sqrt{3}}\right)^2 + \left(\frac{U_{ptol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{pd}}{\sqrt{3}}\right)^2 + \left(\frac{U_{pp}}{\sqrt{3}}\right)^2}$$

Where the sources of uncertainty are

U_{pac}	=	Reference accuracy
U_{pcal}	=	Transmitter calibration
U_{ptemp}	=	Ambient temperature effect
U_{padc}	=	ADC tolerance
U_{ptol}	=	Transmitter calibration tolerance
U_{pd}	=	Drift
U_{pp}	=	Power supply effect

4.1.1.1 Reference accuracy U_{pac}

The reference accuracy is taken from the *product data sheet* of PT model 3051[2] Rosemount for the pressure transducer. From the product technical bulletin, turndown ratio is 2,3:1, this has result

U_{pac} = ± 0,026 % of span for PT - 1423

U_{pac} = ± 0,025 % of span for PT - 1443

[2] *Rosemount Model 3051 Pressure Transmitter*

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.1.2 Transducer calibration U_{pcal}

The pressure transducers are calibrated at site, with all corrections made it is estimated that the uncertainty in measurement is calculated from calibration reports at site and associated with the reference standard uncertainty. So

U_{pcal} = ± 0,033% of span for PT - 1423

U_{pcal} = ± 0,063% of span for PT – 1443

when calculating the standard uncertainty a normal distribution is assumed with a coverage factor k = 2.

4.1.1.3 Ambient temperature effect U_{ptemp}

A contribution is made to the uncertainty budget for variations in ambient temperature, including the variation between the temperature during the calibration of the transducer and the operating temperature. The estimated uncertainty for this is taken from the product data sheet.

For turndown ratio for span greater than 1/10 of URL

U_{ptemp} = ± (0,0125% URL + 0,0625% span) over 28 °C

As the URL of transducer is 137,9 bar and the span is 60 bar, the resulting in an uncertainty of:

U_{ptemp} = ± 0,055 bar/28 °C

The tolerance for the maximum temperature deviation is assumed to be ± 10 °C, therefore the uncertainty due to temperature changes is

U_{ptemp} = ± 0,049 % of reading

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.1.4 ADC tolerance U_{padc}

The analogue to digital converter at the input of the flow computer is checked by routine calibration to be within the tolerance of ± 0,1% of span. However the U_{pdac} after calibration is:

U_{padc} = ± 0,05 % of reading for PT - 1423

U_{padc} = ± 0,04 % of reading for PT - 1443

When calculating the standard uncertainty a rectangular distribution is assumed with a coverage factor k = 1,73.

4.1.1.5 Transducer calibration tolerance U_{ptol}

The acceptance criterion for the calibration of the pressure transducers is equal to the absolute maximum error of test points over calibrated span. Therefore

U_{ptol} = ± 0,050% of span for PT - 1423

U_{ptol} = ± 0,044% of span for PT – 1443

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.1.6 Drift U_{pd}

The contribution due to *drift* in transducer output is calculated from the value for stability taken from the product data sheet and the calibration interval.

Stability = ± 0,125% URL/ 60 months

As the upper range limit of the transducer is 137,9 bar and the calibration interval 1 year, the component assumed for drift is taken to be 0,125 % URL,

U_{pd} = ± 0,034 bar/ year

Once a sufficient number of calibrations have been performed over time this value can be recalculated from the calibration results.

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.1.7 Power supply effect U_{pp}

A contribution is made to the uncertainty budget for variations in power supply, including the variation between the power supply during the calibration of the transmitter and that during operation. The estimated uncertainty for this is taken from the product data sheet.

Power supply effect = < ± 0,005% of calibrated span per volt

The tolerance for the maximum power supply deviation is assumed to be ± 5 volts, therefore

U_{pp} = ± 0,025% of calibrated span

When calculating the standard uncertainty a rectangular distribution is assumed with a coverage factor k = 1,73.

4.1.1.8 Combined uncertainty

The combined standard uncertainty calculated using the equation

$$u_p = \sqrt{\left(\frac{U_{pac}}{\sqrt{3}}\right)^2 + \left(\frac{U_{pcal}}{2}\right)^2 + \left(\frac{U_{ptemp}}{\sqrt{3}}\right)^2 + \left(\frac{U_{padc}}{\sqrt{3}}\right)^2 + \left(\frac{U_{ptol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{pd}}{\sqrt{3}}\right)^2 + \left(\frac{U_{pp}}{\sqrt{3}}\right)^2}$$

is represented in a tabular form as shown below, note that the sensitivity coefficient c_i for each input is 1 as they are all expressed as a percentage of the pressure reading.

- **PT – 1423**

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
Upac	Reference accuracy	0,026	Rectangular	1,73	1,00	0,015	2,34E-04
Upcal	Transmitter calibration	0,061	Normal	2,00	1,00	0,030	9,18E-04
Utemp	Ambient temperature effect	0,059	Rectangular	1,73	1,00	0,034	1,17E-03
Uadc	ADC tolerance (FlowComp. Input)	0,091	Rectangular	1,73	1,00	0,052	2,75E-03
Uptol	Transmitter calibration tolerance	0,091	Rectangular	1,73	1,00	0,052	2,75E-03
Upd	Drift	0,104	Rectangular	1,73	1,00	0,060	3,64E-03
Upp	Power supply effect	0,045	Rectangular	1,73	1,00	0,026	6,89E-04
u(p)	Combined standard uncertainty		Normal			0,110	
U(p)	Expanded uncertainty		Normal (k=2)	2,00		**0,22**	

- **PT – 1443**

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
Upac	Reference accuracy	0,025	Rectangular	1,73	1,00	0,014	2,09E-04
Upcal	Transmitter calibration	0,114	Normal	2,00	1,00	0,057	3,26E-03
Utemp	Ambient temperature effect	0,043	Rectangular	1,73	1,00	0,025	6,21E-04
Uadc	ADC tolerance (FlowComp. Input)	0,091	Rectangular	1,73	1,00	0,052	2,75E-03
Uptol	Transmitter calibration tolerance	0,080	Rectangular	1,73	1,00	0,046	2,11E-03
Upd	Drift	0,209	Rectangular	1,73	1,00	0,121	1,46E-02
Upp	Power supply effect	0,045	Rectangular	1,73	1,00	0,026	6,89E-04
u(p)	Combined standard uncertainty		Normal			0,156	
U(p)	Expanded uncertainty		Normal (k=2)	2,00		**0,31**	

4.1.1.9 The Expanded uncertainty

The expanded uncertainty is calculated by multiplying the combined standard uncertainty by the coverage factor k = 2.

4.1.2 Temperature

Temperature is measured using a RTD PT100. The output is a resistance analogue signal ranged (0 – 80) °C.

The combined standard uncertainty in temperature is estimated by combining inputs derived from the calibration of the transducer and the manufacturer's specification using the following equation:

$$u_t = \sqrt{\left(\frac{U_{tdac}}{\sqrt{3}}\right)^2 + \left(\frac{U_{tda}}{\sqrt{3}}\right)^2 + \left(\frac{U_{tcal}}{2}\right)^2 + \left(\frac{U_{ttemp}}{\sqrt{3}}\right)^2 + \left(\frac{U_{ttol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{td}}{\sqrt{3}}\right)^2 + \left(\frac{U_{tp}}{\sqrt{3}}\right)^2 \left(\frac{U_{tadc}}{\sqrt{3}}\right)^2}$$

where the sources of uncertainty are

U_{tdac}	=	Digital accuracy
U_{tda}	=	D/A Accuracy
U_{tcal}	=	Transducer calibration
U_{ttemp}	=	Ambient temperature effect
U_{ttol}	=	Transducer calibration tolerance
U_{td}	=	Drift
U_{tp}	=	Power supply effect
U_{tadc}	=	ADC tolerance

4.1.2.1 Digital accuracy U_{tdac}

The reference *digital* accuracy is taken from the *product data sheet*[3] for temperature RTD of flowcomputer.

$U_{tdac} = \pm 0{,}1$ °C

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.2.2 D/A accuracy U_{tda}

The reference *D/A* accuracy is taken from the *product data sheet*[3] of RTD

$U_{tda} = \pm 0{,}02\%$ of the calibrated span

The data sheet presents the specifications "conforming to at least 3 σ" therefore when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.2.3 Transducer calibration U_{tcal}

The temperature transmitter are calibrated at lab., with all corrections made it is estimated that the uncertainty in measurement associated with the reference standard , so

U_{tcal} = ± 0,175 % of span for TT 1424

U_{tcal} = ± 0,175 % of span for TT 1444

when calculating the standard uncertainty a normal distribution is assumed with a coverage

[3] *Rosemount Model 3144P Temperature Transmitter*

factor k = 2.

4.1.2.4 Ambient temperature effect U_{ttemp}.

A contribution is made to the uncertainty budget for variations in ambient temperature, including the variation between the temperature during the calibration of the transmitter and the operating temperature. The estimated uncertainty for this is taken from the product data sheet.

U_{ttemp} = ± 0,001% span / 10 °C

The tolerance for the maximum temperature deviation is assumed to be ± 10 °C, therefore:

U_{ttemp} = ± 0,005% of reading

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.2.5 Transducer calibration tolerance U_{ttol}

After calibration at site. The acceptance criterion for the calibration of the temperature transducer is calculated from site reports and has value below:

U_{ttol} = ± 0,038 °C of span for TT 1424

U_{ttol} = ± 0,05 °C of span for TT 1444

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.2.6 Drift U_{td}

The contribution due to *drift* in transmitter output is calculated from the value for stability taken from the tranducer's product data sheet and the calibration interval. This is stated as which ever is the greater of:

Stability = ± 0,1% reading/24 months

Therefore for an annual calibration:

U_{td} = ± 0,05 % of reading/ 12 months

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.2.7 Power supply effect U_{tp}

A contribution is made to the uncertainty budget for variations in power supply, including the variation between the power supply during the calibration of the transducer and that during operation. The estimated uncertainty for this is taken from the product data sheet.

Power supply effect ≤ ± 0,005% of calibrated span per volt

The tolerance for the maximum power supply deviation is assumed to be ± 5 volts, therefore:

U_{tp} = ± 0,18% of reading

when calculating the standard uncertainty a rectangle distribution is assumed with a coverage factor k = 1,73.

4.1.2.8 ADC tolerance U_{padc}

The analogue to digital converter at the input of the flow computer is checked by routine calibration to be within the tolerance of ± 0,1% of span. However the U_{tdac} after calibration is:

U_{tadc} = ± 0,01 % of reading for TT - 1424

U_{tadc} = ± 0,01 % of reading for TT - 1444

When calculating the standard uncertainty a rectangular distribution is assumed with a coverage factor k = 1,73.

4.1.2.9 Combined uncertainty

The combined standard uncertainty calculated using the equation

$$u_t = \sqrt{\left(\frac{U_{tdac}}{\sqrt{3}}\right)^2 + \left(\frac{U_{tda}}{\sqrt{3}}\right)^2 + \left(\frac{U_{tcal}}{2}\right)^2 + \left(\frac{U_{ttemp}}{\sqrt{3}}\right)^2 + \left(\frac{U_{ttol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{td}}{\sqrt{3}}\right)^2 + \left(\frac{U_{tp}}{\sqrt{3}}\right)^2 \left(\frac{U_{tadc}}{\sqrt{3}}\right)^2}$$

is represented in a tabular form as shown below, note that the sensitivity coefficient c_i for each input is 1 as they are all expressed as a percentage of the temperature reading.

- **For TT – 1424**

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
U_{tdac}	Digital accuracy	0,500	Rectangular	1,73	1,00	0,289	8,35E-02
U_{tda}	D/A Accuracy	0,080	Rectangular	1,73	1,00	0,046	2,14E-03
U_{tcal}	Transmitter calibration	0,700	Normal	2,00	1,00	0,350	1,23E-01
U_{ttemp}	Ambient temperature effect	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{ttol}	Transmitter calibration tolerance	0,190	Rectangular	1,73	1,00	0,110	1,20E-02
U_{td}	Drift	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{tp}	Power supply effect	0,100	Rectangular	1,73	1,00	0,058	3,33E-03
U_{tadc}	ADC test	0,040	Rectangular	1,73	1,00	0,023	5,33E-04
$u_{(t)}$	Combined standard uncertainty		Normal			0,475	
$U_{(t)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,95**	

- **For TT – 1444**

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
U_{tdac}	Digital accuracy	0,500	Rectangular	1,73	1,00	0,289	8,35E-02
U_{tda}	D/A Accuracy	0,080	Rectangular	1,73	1,00	0,046	2,14E-03
U_{tcal}	Transmitter calibration	0,700	Normal	2,00	1,00	0,350	1,23E-01
U_{ttemp}	Ambient temperature effect	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{ttol}	Transmitter calibration tolerance	0,250	Rectangular	1,73	1,00	0,144	2,08E-02
U_{td}	Drift	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{tp}	Power supply effect	0,100	Rectangular	1,73	1,00	0,058	3,33E-03
U_{tadc}	ADC test	0,040	Rectangular	1,73	1,00	0,023	5,33E-04
$u_{(t)}$	Combined standard uncertainty		Normal			0,484	
$U_{(t)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,97**	

4.1.2.10 The Expanded uncertainty

The expanded uncertainty is calculated by multiplying the combined standard uncertainty by the coverage factor k = 2.

4.1.3 Gas composition

The composition of the gas is measured by a gas chromatograph which is calibrated by exposure to a certified calibration gas with a composition representative of the gas to be metered. The calibration gas comprises the following components:

Nitrogen	N_2
Carbon Dioxide	CO_2
Methane	C_1
Ethane	C_2
Propane	C_3
nButane	nC_4
iButane	iC_4
nPentane	nC_5
iPentane	iC_5
neoPentane	$neoC_5$
Hexane	nC_6

The combined standard uncertainty for each component is calculated by

$$u_m := \sqrt{\left(\frac{U_{calg}}{2}\right)^2 + \left(\frac{U_{tol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{rep}}{2}\right)^2}$$

where the sources of uncertainty are taken to be

U_{calg} = the uncertainty in the calibration gas

U_{tol} = The tolerance applied during the calibration

U_{rep} = the repeatability of the measurement

The upstream density is calculated from the gas composition measured by a gas chromatograph, the sensitivity coefficients for each of the input quantities are calculated by successively increasing and decreasing each input by the magnitude of its estimated uncertainty and calculating the effect this has on the resulting density as described in the GUM[4]

A temporary variable Z_i is first calculated

$$Z_i = \frac{1}{2}\left[f\left(x_1,..,x_i + u(x_i),...,x_n\right) - f\left(\left(x_1,..,x_i - u(x_i),...,x_n\right)\right)\right]$$

and where the expanded uncertainty is expressed in the form:

$$u_c(y) = \sqrt{\sum_{i=1}^{N} c_i^2 \cdot u^2(x_i)}$$

the value of each sensitivity coefficient c_i taken as $Z_i/u(x_i)$.

4.1.3.1 Sensitivity coefficients for upstream density uncertainty

The uncertainty calculation for each component and the sensitivity coefficients are shown in the table below

The combined standard uncertainty for each component is calculated by:

$$u_m := \sqrt{\left(\frac{U_{calg}}{2}\right)^2 + \left(\frac{U_{tol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{rep}}{2}\right)^2}$$

Where the sources of uncertainty are taken to be:

U_{calg} = Calibration gas uncertainty

U_{tol} = The calibration tolerance

U_{rep} = The gas chromatograph repeatability

Sensitivity coefficient calculation

	Component value	Cal. gas uncertainty (%)	Calibration tolerance (%)	Chromatograph repeatability (%)	Combined uncertainty (%)	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,903861
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,007561

[4] *Guide to the Expression of Uncertainty in Measurement. Section 5*

CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,063737
C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,087[illegible]
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,066172
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,020323
nC_4	**0,0050**	0,3228	0,10	0,02	0,3438	0,018591
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,007778
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,004789
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,008760
P_1 (bar)	**34,01**				0,1206	1,089926
T_1 (°C)	**20,00**				0,9686	-0,091585

4.1.3.2 Uncertainty in upstream density

By using the line pressure, temperature and live gas data from gas chromatograph, the line compressibility factor and also the line density value can be obtained in accordance to AGA 8 calculation.The combined standard uncertainty is calculated as show below

$$U(Rho) = \sqrt{\left(\frac{U_{(p)\exp}}{2}\right)^2 + \left(\frac{U_{(t)\exp}}{2}\right)^2 + \left(\frac{U_Z}{2}\right)^2 + \left(\frac{U_{N_2}}{2}\right)^2 + \left(\frac{U_{CO_2}}{2}\right)^2 + \left(\frac{U_{C1}}{2}\right)^2 + \left(\frac{U_{C2}}{2}\right)^2 + \left(\frac{U_{C3}}{2}\right)^2 + \left(\frac{U_{i-C4}}{2}\right)^2 + \left(\frac{U_{n-C4}}{2}\right)^2 + \left(\frac{U_{i-C5}}{2}\right)^2 + \left(\frac{U_{n-C5}}{2}\right)^2 + \left(\frac{U_{C6+}}{2}\right)^2}$$

- **For FT – 1420**

Symbol	Source of uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
P	Absolute pressure	0,060	Normal	2,00	1,08993	0,03	1,08E-03
T	Line temperature	0,950	Normal	2,00	-0,09158	0,48	1,89E-03
Z	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,90386	0,02	2,30E-04
N	Nitrogen	0,358	Normal	2,00	0,00756	0,18	1,84E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,06374	0,15	9,52E-05
C_2	Ethane	0,433	Normal	2,00	0,08757	0,22	3,60E-04
C_3	Propane	0,313	Normal	2,00	0,06617	0,16	1,07E-04
iC_4	i-Butane	0,358	Normal	2,00	0,02032	0,18	1,32E-05
nC_4	n-Butane	0,344	Normal	2,00	0,01859	0,17	1,02E-05
iC_5	i-Pentane	0,583	Normal	2,00	0,00778	0,29	5,14E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00479	0,34	2,60E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00876	0,52	2,1[illegible]5
$u_{(Rho)}$	Combined standard uncertainty		Normal			0,080	
$U_{(Rho)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,16**	

- **For FT- 1440**

Symbol	Source of uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
P	Absolute pressure	0,121	Normal	2,00	1,08993	0,06	4,32E-03
T	Line temperature	0,969	Normal	2,00	-0,09158	0,48	1,97E-03
Z	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,90386	0,02	2,30E-04
N	Nitrogen	0,358	Normal	2,00	0,00756	0,18	1,84E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,06374	0,15	9,52E-05
C_2	Ethane	0,433	Normal	2,00	0,08757	0,22	3,60E-04
C_3	Propane	0,313	Normal	2,00	0,06617	0,16	1,07E-04
iC_4	i-Butane	0,358	Normal	2,00	0,02032	0,18	1,32E-05
nC_4	n-Butane	0,344	Normal	2,00	0,01859	0,17	1,02E-05
iC_5	i-Pentane	0,583	Normal	2,00	0,00778	0,29	5,14E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00479	0,34	2,60E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00876	0,52	2,10E-05
$u_{(Rho)}$	Combined standard uncertainty		Normal			0,098	
$U_{(Rho)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,20**	

4.2 Standard density

The standard density is calculated from the gas composition measured by a gas chromatograph.

The functional relationship used to determine the standard density is defined in ISO 6976[5] - 1995. This can be written in the form

$$y = f(x_1, x_2, x_3 \ldots x_i)$$

where the output y (standard density) is a function of a number of inputs x (components forming the gas composition)

The sensitivity coefficients for each of the input quantities are calculated by the same finite difference method as used when calculating the sensitivity coefficients for the upstream density.

4.2.1 Sensitivity coefficients for standard density uncertainty

The uncertainty calculation for each component and the sensitivity coefficients are shown in the table below

The combined standard uncertainty for each component is calculated by

$$u_m := \sqrt{\left(\frac{U_{calg}}{2}\right)^2 + \left(\frac{U_{tol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{rep}}{2}\right)^2}$$

[5] *ISO 6976 Natural Gas – Calculation of calorific values, density, relative density and Wobbe index from a gas composition*

where the sources of uncertainty are taken to be

U_{calg} = Calibration gas uncertainty

U_{tol} = The calibration tolerance

U_{rep} = The gas chromatograph repeatability

The uncertainty calculation for each component and the sensitivity coefficients are shown in the table below

Sensitivity coefficient calculation

	Component value	Cal. gas uncertainty (%)	Calibration tolerance (%)	Chromatograph repeatability (%)	Combined uncertainty (%)	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,763455
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,007249
CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,057793
C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,072299
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,054761
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,01[illegible]
nC_4	**0,0050**	0,3228	0,10	0,02	0,3438	0,015451
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,006486
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,003969
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,007093

4.2.2 Uncertainty in standard density

The ultrasonic meter measured line volume is corrected to base condition in order to obtain the standard volume flow rate at base condition. The correction is done in accordance with PTZ methodology. The PTZ formula is described in equation 4.

The gas compressibility Z and gas density at base condition is determined in flow computer in accordance with ISO 6976 - 1995 with the gas data input from gas chromatograph.

$$u_{(Rhostd)} = \sqrt{\left(\frac{U_Z}{2}\right)^2 + \left(\frac{U_{N_2}}{2}\right)^2 + \left(\frac{U_{CO_2}}{2}\right)^2 + \left(\frac{U_{C1}}{2}\right)^2 + \left(\frac{U_{C2}}{2}\right)^2 + \left(\frac{U_{C3}}{2}\right)^2 + \left(\frac{U_{i_C4}}{2}\right)^2 + \left(\frac{U_{n_C4}}{2}\right)^2 + \left(\frac{U_{i_C5}}{2}\right)^2 + \left(\frac{U_{n_C5}}{2}\right)^2 + \left(\frac{U_{C6+}}{2}\right)^2}$$

The combined standard uncertainty is calculated as shown below

Symbol	Source of uncertainty	Value %	Probability distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,76345	0,02	1,64E-04
N	Nitrogen	0,358	Normal	2,00	0,00725	0,18	1,69E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,05779	0,15	7,83E-05
C_2	Ethane	0,433	Normal	2,00	0,07230	0,22	2,45E-04
C_3	Propane	0,313	Normal	2,00	0,05476	0,16	7,36E-05
iC_4	i-Butane	0,358	Normal	2,00	0,01683	0,18	9,06E-06
nC_4	n-Butane	0,344	Normal	2,00	0,01545	0,17	7,05E-06
iC_5	i-Pentane	0,583	Normal	2,00	0,00649	0,29	3,58E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00397	0,34	1,79E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00709	0,52	1,38E-05
$u_{(Rhostd)}$	Combined standard uncertainty		Normal			0,056	
$U_{(Rhostd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,11**	

4.3 The Combined uncertainty in standard volume flow

With references to equation 2 and 4 the uncertainty equation of standard volume flow for both elements is, therefore

$$u_{qstd} = \sqrt{\left(\frac{U_m}{2}\right)^2 + \left(\frac{U_{(Rho)\exp}}{2}\right)^2 + \left(\frac{U_{(Rhostd)}}{2}\right)^2 + \left(\frac{U_{(fc)\exp}}{\sqrt{3}}\right)^2}$$

with an assumption of normal distribution applied for the uncertainty

- **For FT – 1420**

Symbol	Source of uncertainty	Value %	Probability distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
$u_{(m)}$	Ultrasonic calibration	0,298	Normal	2,00	1,00	0,15	0,022
$u_{(Rho)}$	Relation density	0,159	Normal	2,00	1,00	0,08	0,006
$u_{(Rhostd)}$	Standard density	0,111	Normal	2,00	1,00	0,06	0,003
$u_{(Fc)}$	Flow computer calculation	0,001	Normal	2,00	1,00	0,00	2,50E-07
$u_{(Qstd)}$	Combined standard uncertainty		Normal			0,178	
$U_{(Qstd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,36**	

- **For FT – 1440**

Symbol	Source of uncertainty	Value %	Probability distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
$u_{(m)}$	Ultrasonic calibration	0,304	Normal	2,00	1,00	0,15	0,023
$u_{(Rho)}$	Relation density	0,196	Normal	2,00	1,00	0,10	0,010
$u_{(Rhostd)}$	Standard density	0,111	Normal	2,00	1,00	0,06	0,003
$u_{(FC)}$	Flow computer calculation	0,001	Normal	2,00	1,00	0,00	2,50E-07
$u_{(Qstd)}$	Combined standard uncertainty		Normal			0,189	
$U_{(Qstd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,38**	

The normalized volume flow expanded uncertainty for both ultrasonic meters under various flow conditions is given in the following tables

- **For FT – 1420**

% Q_{max}	$U_{(m)}$	$U_{(Rho)}$	$U_{(Rho.std)}$	$U_{(FC)}$	$U_{(Qstd)}$	$U_{Exp.std.fr}$
5,0	0,29	0,16	0,11	0,001	0,17	**0,35**
10,1	0,30	0,16	0,11	0,001	0,18	**0,36**
25,2	0,30	0,16	0,11	0,001	0,18	**0,36**
39,9	0,29	0,16	0,11	0,001	0,17	**0,35**
69,8	0,28	0,16	0,11	0,001	0,17	**0,34**
80,4	0,29	0,16	0,11	0,001	0,17	**0,35**
Distribution	Normal	Normal	Normal	Normal		Normal
Divisor	2	2	2	2		2
ci	1	1	1	1		

- **For FT – 1440**

% Q_{max}	$U_{(m)}$	$U_{(Rho)}$	$U_{(Rho.std)}$	$U_{(FC)}$	$U_{(Qstd)}$	$U_{Exp.std.fr}$
5,0	0,28	0,20	0,11	0,001	0,18	**0,36**
10,0	0,28	0,20	0,11	0,001	0,18	**0,36**
25,0	0,30	0,20	0,11	0,001	0,19	**0,37**
40,3	0,30	0,20	0,11	0,001	0,19	**0,38**
69,8	0,29	0,20	0,11	0,001	0,18	**0,37**
79,8	0,30	0,20	0,11	0,001	0,19	**0,37**
Distribution	Normal	Normal	Normal	Normal		Normal
Divisor	2	2	2	2		2
ci	1	1	1	1		

The uncertainty in the energy flowrate can be calculated by including the uncertainty in the calorific value E_{CV}.

$$u_{qe} = \sqrt{\left(\frac{U_{(qstd)\exp}}{2}\right)^2 + \left(\frac{U_{(cv)\exp}}{2}\right)^2}$$

5.1 Calorific value

The calorific value (CV) is calculated from the gas composition measured by a gas chromatograph.

The functional relationship used to determine the calorific value is defined in ISO 6976[6]. this can be written in the form

$$y = f(x_1, x_2, x_3 \ldots x_i)$$

where the output y (CV) is a function of a number of inputs x (components forming the gas composition)

The sensitivity coefficients for each of the input quantities are calculated by the same finite difference method as used when calculating the sensitivity coefficients for the upstream density.

5.1.1 Sensitivity coefficients for calorific value uncertainty

The uncertainty calculation for each component and the sensitivity coefficients are shown in the table below

The combined standard uncertainty for each component is calculated by

$$u_m := \sqrt{\left(\frac{U_{calg}}{2}\right)^2 + \left(\frac{U_{tol}}{\sqrt{3}}\right)^2 + \left(\frac{U_{rep}}{2}\right)^2}$$

where the sources of uncertainty are taken to be:

U_{calg} = Calibration gas uncertainty

U_{tol} = The calibration tolerance

U_{rep} = The gas chromatograph repeatability

The uncertainty calculation for each component and the sensitivity coefficients are shown in the table below:

	Value (%)	Composition uncertainty (%)	Chromatograph tolerance (%)	Chromatograph repeatability (%)	Combined uncertainty (%)	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,829712
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,000009
CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,000191

[6] *ISO 6976 Natural Gas – Calculation of calorific values, density, relative density and Wobbe index from a gas composition.*

C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,073476
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,05[illegible]8
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,016265
nC_4	**0,0050**	0,3228	0,10	0,02	0,3438	0,014980
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,006213
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,003810
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,006763

5.1.2 Uncertainty in calorific value

From the experiment of 5 consecutive cycles of validation check of the GC with using calibration gas, to composition uncertainty is determined in accordance with ISO 6976 guideline. The combined uncertainty of heating value at base condition is given as follow

$$u_{(cv)} = \sqrt{\left(\frac{U_{ISO6976}}{2}\right)^2 + \left(\frac{U_{N_2}}{2}\right)^2 + \left(\frac{U_{CO_2}}{2}\right)^2 + \left(\frac{U_{C1}}{2}\right)^2 + \left(\frac{U_{C2}}{2}\right)^2 + \left(\frac{U_{C3}}{2}\right)^2 + \left(\frac{U_{i_C4}}{2}\right)^2 + \left(\frac{U_{n_C4}}{2}\right)^2 + \left(\frac{U_{i_C5}}{2}\right)^2 + \left(\frac{U_{n_C5}}{2}\right)^2 + \left(\frac{U_{C6+}}{2}\right)^2}$$

The combined standard uncertainty is calculated as shown below

Symbol	Source of uncertainty	Value %	Probability distribution	Divisor / Multiplier	c_i	u_i	$(c_i \cdot u_i)^2$
	Equation of state	0.10	Normal	2.00	1.00000	0,05	2,50E-03
C_1	Methane	0.03	Normal	2.00	0.82971	0.02	1.94E-04
N	Nitrogen	0,36	Normal	2.00	0.00001	0,18	2.44E-12
CO_2	Carbon Dioxide	0,31	Normal	2.00	0.00019	0,15	8.54E-10
C_2	Ethane	0.43	Normal	2.00	0.07348	0.22	2,53E-04
C_3	Propane	0.31	Normal	2.00	0.05397	0.16	7.14E-05
iC_4	i-Butane	0.36	Normal	2.00	0.01627	0.18	8.46E-06
nC_4	n-Butane	0,34	Normal	2.00	0.01498	0,17	6,63E-06
iC_5	i-Pentane	0.58	Normal	2.00	0.00621	0,29	3.28E-06
nC_5	n-Pentane	0.67	Normal	2.00	0.00381	0.34	1.65E-06
nC_6	n-Hexane	1.05	Normal	2.00	0.00676	0,52	1.25E-05
$u_{(CV)}$	Combined standard uncertainty		Normal			0,055	
$U_{(CV)}$	Expanded uncertainty		Normal (k=2)	2.00		**0,11**	

5.2 Total uncertainty in energy flowrate

With applying a mathematical deviation on equation above, uncertainty elements of energy flow is derived as follow

$$u_{qc} = \sqrt{\left(\frac{U_{(qstd)\exp}}{2}\right)^2 + \left(\frac{U_{(cv)\exp}}{2}\right)^2}$$

- **For FT – 1420**

Symbol	Source of uncertainty	Value %	Probability distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
q_{std}	Standard flowrate	0,356	Normal	2,00	1,00	0,18	0,032
CV	Calorific value	0,110	Normal	2,00	1,00	0,06	0,003
$u_{(E)}$	Combined standard uncertainty		Normal			0,186	
$U_{(E)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,37**	

- **For FT – 1440**

Symbol	Source of uncertainty	Value %	Probability distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
q_{std}	Standard flowrate	0,379	Normal	2,00	1,00	0,19	0,036
CV	Calorific value	0,110	Normal	2,00	1,00	0,06	0,003
$u_{(E)}$	Combined standard uncertainty		Normal			0,197	
$U_{(E)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,39**	

The calculations summarised in the following table is performed at differential flow range increments throughout the operating range of the flowrate of USM

- **For FT –1420**

Ultrasonic meter flow range (%)	Standard uncertainty (%)		Combined standard (%)	Expanded (%) k = 2
	U_{qstd}	U_{cv}		
5,0	0,35	0,11	0,18	**0,36**
10,1	0,36	0,11	0,19	**0,37**
25,2	0,36	0,11	0,19	**0,37**
39,9	0,35	0,11	0,18	**0,36**
69,8	0,34	0,11	0,18	**0,36**
80,4	0,35	0,11	0,18	**0,36**
Divisor	2	2		
Ci	1	1		

- **For FT – 1440**

Ultrasonic meter flow range (%)	Standard uncertainty (%)		Combined standard (%)	Expanded (%) k = 2
	U_{qstd}	U_{cv}		
5,0	0,36	0,11	0,19	**0,38**
10,0	0,36	0,11	0,19	**0,37**
25,0	0,37	0,11	0,19	**0,39**
40,3	0,38	0,11	0,20	**0,39**
69,8	0,37	0,11	0,19	**0,38**
79,8	0,37	0,11	0,19	**0,39**
Divisor	2	2		
Ci	1	1		

6.0 SUMMARY OF METERING SYSTEM UNCERTAINTY

This section provides the summary results of the uncertainty estimation for each gas metering system. The detail calculation has been performed based on ISO5168- 2002 and shown in Appendix A, B and C.

The uncertainty estimation performed has been based on results described in the product catalogue, results obtained from some analysis and results from some third party testing facility

In summary, the uncertainty estimation was within the project allowable limit. The summary result was indicated as below

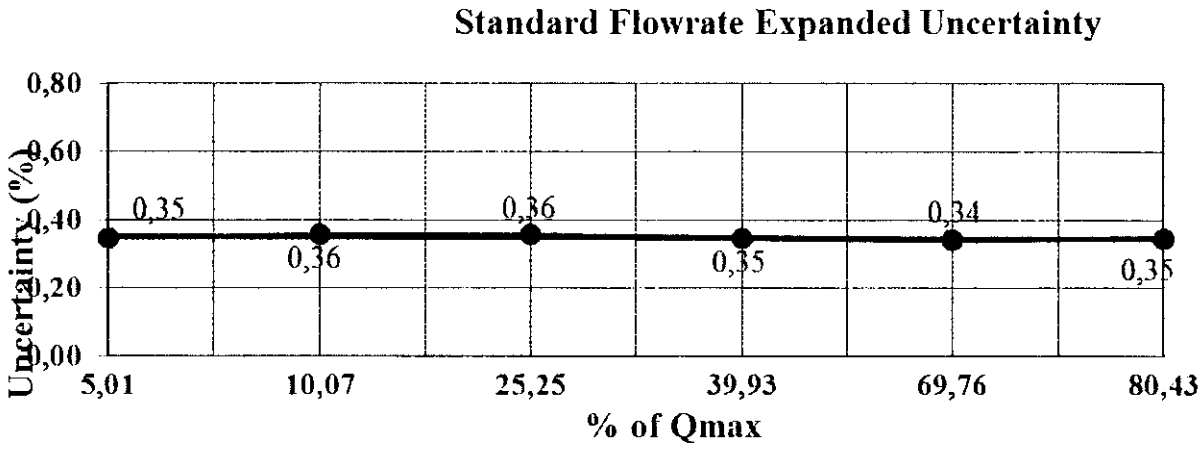
- **For FQI – 1420**

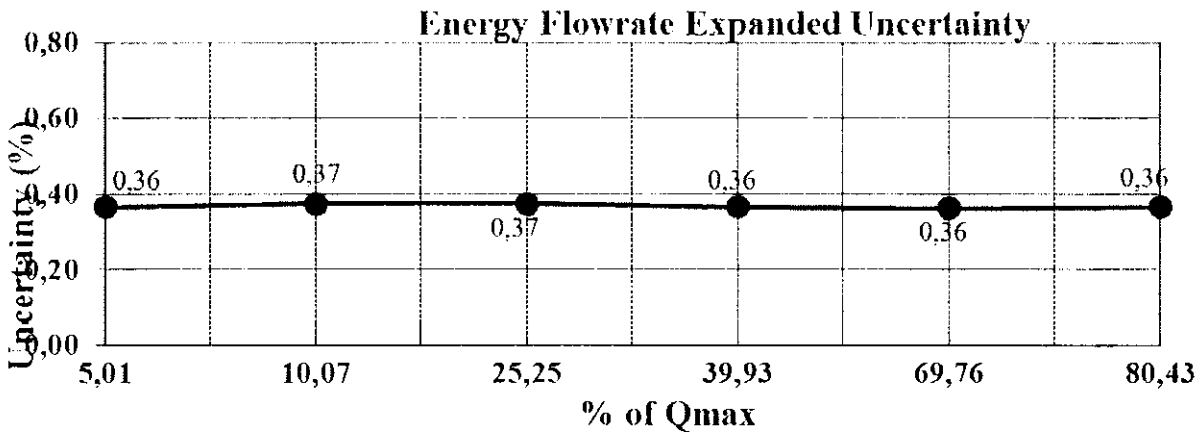
  The maximum expanded uncertainty for standard volume flow rate = 0,36 %

  The maximum expanded uncertainty for energy flow rate = 0,37 %

% Q_{max}	Flowrate (m^3/h)	Standard flowrate (m^3/h) *)	Energy flowrate (MJ/h)	U_{exp} standard flowrate (%)	U_{exp} energy flowrate (%)
5,0	**250,31**	**8817,9**	**358578,0**	**0,35**	**0,36**
10,1	**503,54**	**17738,8**	**721338,9**	**0,36**	**0,37**
25,2	**1262,44**	**44473,4**	**1808490,1**	**0,36**	**0,37**
39,9	**1996,41**	**70329,7**	**2859928,2**	**0,35**	**0,36**
69,8	**3488,23**	**122883,7**	**4997013,2**	**0,34**	**0,36**
80,4	**4021,72**	**141677,6**	**5761256,6**	**0,35**	**0,36**

*) *The unit of volume flowrate is converted to the reference condition at 1,01325 bar a and 15 °C*





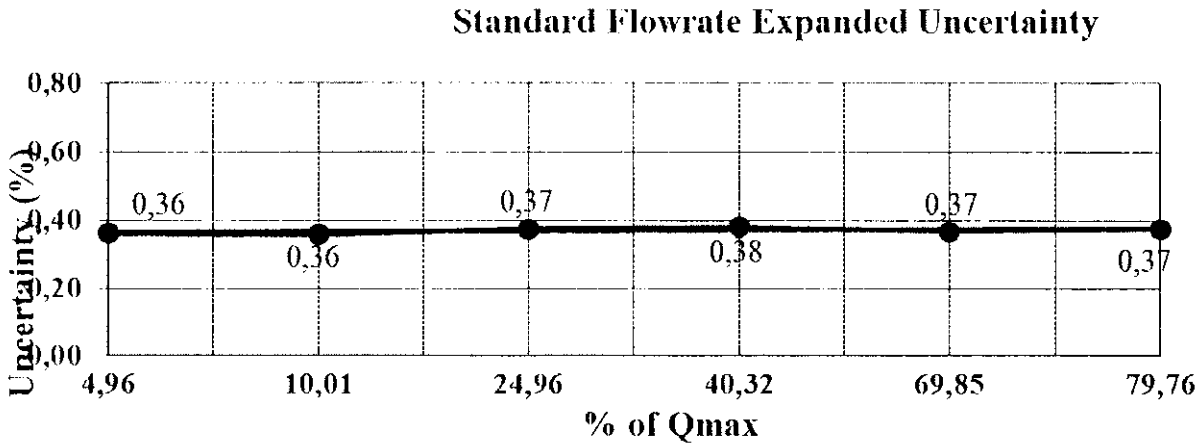
- **For FQI – 1440**

The maximum expanded uncertainty for standard volume flow rate = 0,39 %

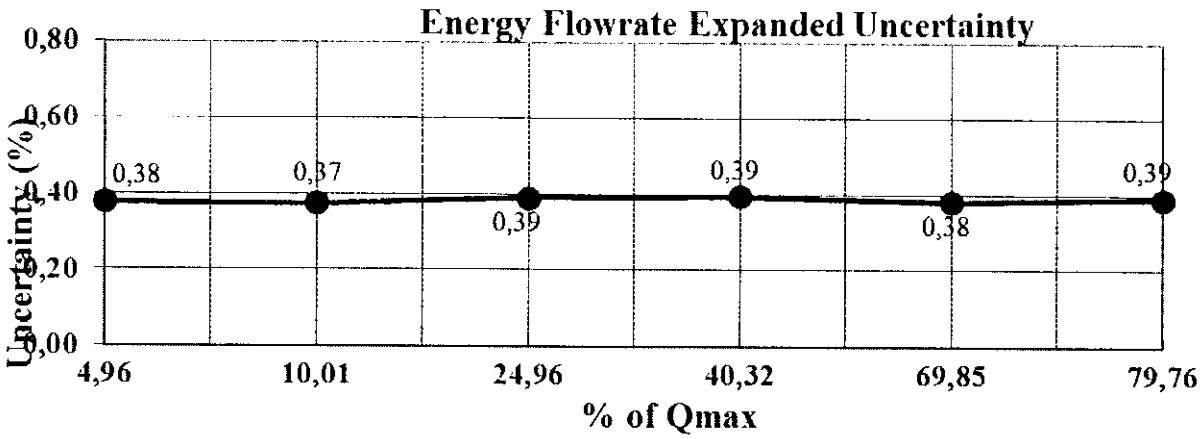
The maximum expanded uncertainty for energy flow rate = 0,38%

% Q_{max}	Flowrate (m^3/h)	Standard flowrate (m^3/h) *)	Energy flowrate (MJ/h)	U_{exp} standard flowrate (%)	U_{exp} energy flowrate (%)
5,0	**247,79**	**8729,2**	**354968,0**	**0,36**	**0,38**
10,0	**500,27**	**17623,6**	**716654,5**	**0,36**	**0,37**
25,0	**1247,77**	**43956,6**	**1787474,8**	**0,37**	**0,39**
40,3	**2015,98**	**71019,2**	**2887962,9**	**0,38**	**0,39**
69,8	**3492,27**	**123026,1**	**5002800,7**	**0,37**	**0,38**
79,8	**3987,93**	**140487,2**	**5712851,2**	**0,37**	**0,39**

*) *The unit of volume flowrate is converted to the reference condition at 1,01325 bar a and 15 °C*



6.0 SUMMARY OF METERING SYSTEM UNCERTAINTY



7.0 REFERENCES

1. BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, OIML, *Guide to the Expression of Uncertainty in Measurement*. International Organisation for Standardisation, Geneva, Switzerland.

2. AGA 8:1994 - Compressibility factor of natural gas and other related hydrocarbon gases

3. AGA 9 Measurement of Gas by Multipath Ultrasonic Meters

4. International Standard ISO 6976-1994 Natural Gas - Calculation of calorific values, density, relative density and Wobbe index from a gas composition.

5. Rosemount Model 3051 Pressure Transmitter, *Product Data Sheet* - Emerson Process Management.

6. Rosemount Model 3144P Temperature Transmitter, *Product Data Sheet* - Emerson Process Management.

7. ISO 5168:2005 Measurement of fluid flow – Evaluation of uncertainty.

QUATEST 3®

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UNCERTAINTY CALCULATION
STREAM FQI-1420

2020

PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

Uncertainty Summary Report

ISO 5168 - 2005

AGA8-94 (N - using AGA8-85)	Y
ISO 6976-95 (N - using ISO6976-83)	Y

Measurements

Line Pressure	34,01	bar a
Line Temperature	20,00	°C

Pressure PT-1423

Static

Manufacturer	Rosemount
Model No	3051CG5A02
Upper range limit	137,9 bar
Span	60,0 bar
SN	RS 0911811

Composition

	% mole	Cal Gas Accuracy	Cal Gas Accuracy (%)	GC Meas (%Mol)	% tolerance
Methane (C_1)	88,7970%	0,0210	0,0236	88,8070	0,0113
Nitrogen (N)	0,4849%	0,0017	0,3506	0,4846	0,0619
Carbon Dioxide (CO_2)	2,4555%	0,0071	0,2891	2,4576	0,0855
Ethane (C_2)	4,4840%	0,0170	0,3791	4,4759	0,1806
Propane (C_3)	2,3161%	0,0069	0,2979	2,3180	0,0820
Water (H_2O)	0,0000%	0,0000	0,0000	0,0000	0,0000
Hydrogen Sulphide (H_2S)	0,0000%	0,0000	0,0000	0,0000	0,0000
Hydrogen (H_2)	0,0000%	0,0000	0,0000	0,0000	0,0000
Carbon Monoxide (CO)	0,0000%	0,0000	0,0000	0,0000	0,0000
Oxigen (O_2)	0,0000%	0,0000	0,0000	0,0000	0,0000
i-Butane (iC_4)	0,5400%	0,0018	0,3333	0,5406	0,1111
n-Butane (nC_4)	0,4956%	0,0016	0,3228	0,4961	0,1009
i-Pentane (iC_5)	0,1676%	0,00094	0,5608	0,1674	0,1372
n-Pentane (nC_5)	0,1025%	0,0006	0,5950	0,1028	0,2731
n-Hexane (nC_6)	0,1534%	0,0016	1,0430	0,1535	0,0652
n-Heptane (nC_7)	0,0000%		0,0000	0,0000	0,0000
n-Octane (nC_8)	0,0000%		0,0000	0,0000	0,0000
n-Nonane (nC_9)	0,0000%		0,0000	0,0000	0,0000
n-Decane (nC_{10})	0,0000%		0,0000	0,0000	0,0000
Helium (He)	0,0000%		0,0000	0,0000	0,0000
Argon (Ar)	0,0000%		0,0000	0,0000	0,0000
Sum	99,9967%				

Chromatograph repeatability 0,02 %

	Static	
Reference Accuracy	0,026	%span
Transmitter calibration	0,033	% span
Ambient temperature effect		
%URL	0,0125	% URL
%Span	0,0625	% span
at	28	°C
with max dev temperature	10	°C
Transm. Cal. Tolerance	0,050	% span
Drift (stablity)	0,125	% URL
with stable duration	60	months
and calibration interval	12	months
ADC Tolerance	0,050	% span
Supply voltage effect		
Less than,% span	0,005	% span
Voltage variation	5,0	V

Equation of state **Uncertainty of correct factor**

Compresibility factor, AGA8	0,1 %
Standard density, ISO6976	0,1 %
Caloritic value, ISO6976	0,1 %

Uncertainty Limit 1 %

Temperature TT-1424

	Transmiter	RTD
Manufacturer	Rosemount	Rosemount
Model	3144P D1A	R68
Type	RTD	PT100
Span	80 °C	(-200 - 400)
SN	12025577	

	Trans	
Digital Accuracy	0,1	°C
D/A Accuracy	0,02	% span
Transmitter Calibration Tolerance	0,038	°C
Ambient temperature effect		
with Max temp.deviation	10	°C
Digital accuracy (°C / °C)	0,0015	°C/°C
D/A effect (%span / °C)	0,01	% span /10 °C
Drift (stablity)	0,1	% reading
with stable duration	24	months
and calibration interval	12	months
Transmitter calibration	0,175	% span
Power supply effect	0,005	% span/volt
with voltage variation	5	v
ADC Tolerance	0,01	% span

FlowComputer	FQI-1420
Manufacturer	Daniel
Model	FloBoss S600
SN	NA
ADC resolution	32 bit
ADC accuracy	- %
Input ressistor	- %
Calculation accuracy	0,001 %

USM/TB	FT-1420
Manufacturer	Daniel
Meter type	Ultrasonic
Model	3400
G-rating	
SN	02-050175
Meter dia.	10 inches
Maximum Flowrate	5000 m3/h
Maximum Pressure	80
Impulses / m^3 (HF)	3582,21
Impulses / m^3 (LF)	3582,21

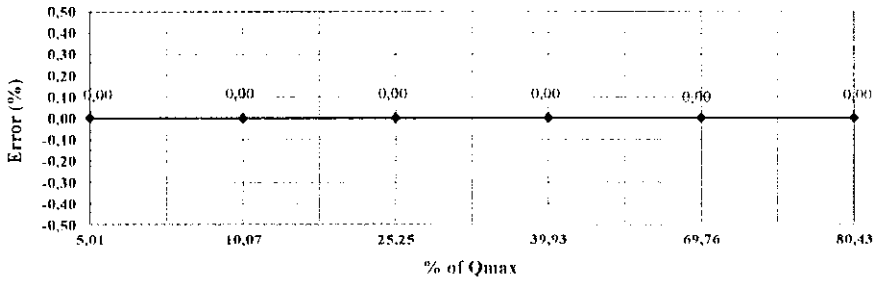
% Qmax	Flowrate (m^3/h)	U_{cal} (%)	U_{lin} (%)	U_{rpt} (%)
5,0	250,31	0,15	0,000	0,30
10,1	503,54	0,17	0,000	0,30
25,2	1262,44	0,17	0,000	0,30
39,9	1996,41	0,15	0,000	0,30
69,8	3488,23	0,14	0,000	0,30
80,4	4021,72	0,15	0,000	0,30
		(Certificate)	(Certificate)	(AGA 9)

PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

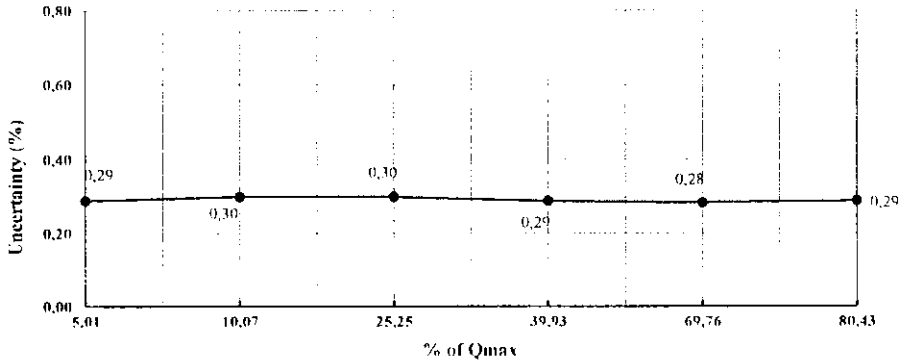
Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
U_{cal}	Ultrasonic calibration	0,170	Normal	2,00	1,00	0,09	7,23E-03
U_{tol}	Ultrasonic calibration tolerance	0,000	Rectangular	1,73	1,00	0,00	0,00E+00
U_{se}	System effect	0,300	Triangular	2,45	1,00	0,12	1,50E-02
u(M)	Combined standard uncertainty		Normal			0,149	
U(M)	Expanded uncertainty		Normal (k=2)	2,00		**0,30**	

% Qmax	Q(m³/h)	Deviation (%)	U_{cal} (%)	U_{tol} (%)	U_{se} (%)	U_r (%)	$U_{exp(M)}$ (%)
5,0	250,3	0,000	0,15	0,000	0,30	0,14	0,29
10,1	503,5	0,000	0,17	0,000	0,30	0,15	0,30
25,2	1262,4	0,000	0,17	0,000	0,30	0,15	0,30
39,9	1996,4	0,000	0,15	0,000	0,30	0,14	0,29
69,8	3488,2	0,000	0,14	0,000	0,30	0,14	0,28
80,4	4021,7	0,000	0,15	0,000	0,30	0,14	0,29

Error Curve



USM Calibration



PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
Upac	Reference accuracy	0,026	Rectangular	1,73	1,00	0,015	2,34E-04
Upcal	Transmitter calibration	0,061	Normal	2,00	1,00	0,030	9,18E-04
Utemp	Ambient temperature effect	0,059	Rectangular	1,73	1,00	0,034	1,17E-03
Uadc	ADC tolerance (FlowComp. Input)	0,091	Rectangular	1,73	1,00	0,052	2,75E-03
Uptol	Transmitter calibration tolerance	0,091	Rectangular	1,73	1,00	0,052	2,75E-03
Upd	Drift	0,104	Rectangular	1,73	1,00	0,060	3,64E-03
Upp	Power supply effect	0,045	Rectangular	1,73	1,00	0,026	6,89E-04
u(p)	Combined standard uncertainty		Normal			0,110	
U(p)	Expanded uncertainty		Normal (k=2)	2,00		**0,22**	

URL	137,9	bar
Span	60	bar
Reading	33,00	bar
Turndown	2,3 :1	

Reference Accuracy

	0,026	% span
=	0,016	bar
=	0,048	% reading

Transmitter calibration

Cert. of calibration	0,033	% span
=	0,020	bar
=	0,061	% reading

Ambient temperature effect

Maximum temperature deviation		10	°C
		28	°C
0,0125	% of URL	= 0,017	bar/28°C
0,0625	% of span	= 0,038	bar/28°C
	sum =	0,055	bar/28°C
temperature effect	=	0,020	bar
	=	0,059	% reading

ADC test

Cert. of calibration	0,050	% span
=	0,030	bar
=	0,091	% reading

Transmitter Calibration Tolerance

Cert. of calibration	0,050	% span
=	0,030	bar
=	0,091	% reading

Drift

Stability Duration	60	months
Calibration interval	12	months
Stability =	0,125	% URL/60 months
=	0,025	% URL/12 months
=	0,034	bar/12 months
=	0,034	bar
=	0,104	% reading

Power supply effect

Voltage variation	5	Volts
Stability =	0,005	% span/volt
=	0,003	bar/volt
=	0,015	bar
=	0,045	% reading

PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
U_{tdac}	Digital accuracy	0,500	Rectangular	1,73	1,00	0,289	8,35E-02
U_{tda}	D/A Accuracy	0,080	Rectangular	1,73	1,00	0,046	2,14E-03
U_{tcal}	Transmitter calibration	0,700	Normal	2,00	1,00	0,350	1,23E-01
U_{ttemp}	Ambient temperature effect	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{ttol}	Transmitter calibration tolerance	0,190	Rectangular	1,73	1,00	0,110	1,20E-02
U_{td}	Drift	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{tp}	Power supply effect	0,100	Rectangular	1,73	1,00	0,058	3,33E-03
U_{tadc}	ADC test	0,040	Rectangular	1,73	1,00	0,023	5,33E-04
$u_{(t)}$	Combined standard uncertainty		Normal			0,475	
$U_{(t)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,95**	

Model	3144P D1A		
Type	RTD		
Span	80	°C	
Reading	20	°C	
	– 295,2	K	

Digital Accuracy

From data sheet: 0,1 °C
– 0,50 % reading

D/A Accuracy

From data sheet: 0,02 % span
0,02 °C
– 0,08 % reading

Transmitter Calibration Tolerance

Cert. of calibration
– 0,04 °C
– 0,1900000 % reading

Ambient temperature effect

Maximum temperature deviation: 10 °C
D/A effect (0.001%span / °C): 0,010 °C
– 0,0500000 % reading
0,010000 °C
Ambient temperature effect: 0,0500000 % reading

Drift

Calibration interval: 12 months
Stability Duration: 21 months
Stability: greater of 0,1 % reading/24 months
or 0,05 % reading/12 months

Transmitter calibration

Cet. Of calibration: 0,18 % span
0,14 °C
– 0,70 % reading

Power supply effect

Voltage variation: 5 Volts

Stability
– 0,005 % span/volt
– 0,004 °C/volt
– 0,02 °C
– 0,1000 % reading

ADC test

Cet. Of calibration: 0,010 % span
0,008 °C
– 0,040 % reading

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,76345	0,02	1,64E-04
N	Nitrogen	0,358	Normal	2,00	0,00725	0,18	1,69E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,05779	0,15	7,83E-05
C_2	Ethane	0,433	Normal	2,00	0,07230	0,22	2,45E-04
C_3	Propane	0,313	Normal	2,00	0,05476	0,16	7,36E-05
iC_4	i-Butane	0,358	Normal	2,00	0,01683	0,18	9,06E-06
nC_4	n-Butane	0,344	Normal	2,00	0,01545	0,17	7,05E-06
iC_5	i-Pentane	0,583	Normal	2,00	0,00649	0,29	3,58E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00397	0,34	1,79E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00709	0,52	1,38E-05
$u_{(Rhostd)}$	Combined standard uncertainty		Normal			0,056	
$U_{(Rhostd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,11**	

Standard density calculated in accordance with ISO6976-1995

Sensitivity coefficient calculation

	Value	Cal gas Uncertainty %	Calibration Tolerance %	Chromat Repeatability %	Combined Uncertainty %	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,763455
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,007249
CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,057793
C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,072299
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,054761
H_2O	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2S	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
CO	0,0000	0,0000	0,00	0,00	0,0000	0,000000
O_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,016830
nC_4	**0,0050**	0,3228	0,10	0,[illegible] Calculat	0,3438	0,015451
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,006486
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,003969
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,007093
nC_7	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_8	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_9	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_{10}	0,0000	0,0000	0,00	0,00	0,0000	0,000000
He	0,0000	0,0000	0,00	0,00	0,0000	0,000000
Ar	0,0000	0,0000	0,00	0,00	0,0000	0,000000

Std Rho kg/m^3 **0,795579** **0,0489%**

Tb, °C	15	
Pb, Bar	1,01325	
Tz, °C	15	15
Thv, °C	15	0
Hs.MJ/Sm^3.10^{-5}	40,6645618	42,842891
R den	0,64923528	0,6491983
M mass. Kg/kmol	18,7609057	18,760498
Std.Rho., kg/Sm^3	0,79557921	0,7955767
Z	0,9973135	
Hs.kJ/kg	51,1131528	
Hs. kJ/mol	958,929039	
Hi.MJ/Sm^3.10^{-5}	36,7417527	

PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
P	Absolute pressure	0,060	Normal	2,00	1,08993	0,03	1,08E-03
T	Line temperature	0,950	Normal	2,00	-0,09158	0,48	1,89E-03
Z	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,90386	0,02	2,30E [illegible]
N	Nitrogen	0,358	Normal	2,00	0,00756	0,18	1,84E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,06374	0,15	9,52E-05
C_2	Ethane	0,433	Normal	2,00	0,08757	0,22	3,60E-04
C_3	Propane	0,313	Normal	2,00	0,06617	0,16	1,07E-04
iC_4	i-Butane	0,358	Normal	2,00	0,02032	0,18	1,32E-05
nC_4	n-Butane	0,344	Normal	2,00	0,01859	0,17	1,02E-05
iC_5	i-Pentane	0,583	Normal	2,00	0,00778	0,29	5,14E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00479	0,34	2,60E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00876	0,52	2,10E-05
u(Z1)	Combined standard uncertainty		Normal			0,080	
U(Z1)	Expanded uncertainty		Normal (k=2)	2,00		**0,16**	

Density calculated in accordance with AGA8 1992

Sensitivity coefficient calculation

		Component Value	Cal gas Uncertainty %	Calibration Tolerance %	Chromat Repeatability %	Combined Uncertainty %	c_i
C_1		**0,8880**	0,0236	0,01	0,02	0,0336	0,903861
N		**0,0048**	0,3506	0,06	0,02	0,3584	0,007561
CO_2		**0,0246**	0,2891	0,09	0,02	0,3062	0,063737
C_2		**0,0448**	0,3791	0,18	0,02	0,4332	0,087573
C_3		**0,0232**	0,2979	0,08	0,02	0,3133	0,066172
H_2O		0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2S		0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2		0,0000	0,0000	0,00	0,00	0,0000	0,000000
CO		0,0000	0,0000	0,00	0,00	0,0000	0,000000
O_2		0,0000	0,0000	0,00	0,00	0,0000	0,000000
iC_4		**0,0054**	0,3333	0,11	0,02	0,3577	0,020323
nC_4		**0,0050**	0,3228	0,10	0,02	0,3438	0,018591
iC_5		**0,0017**	0,5608	[illegible]	0,02	0,5831	0,007778
nC_5		**0,0010**	0,5950	0,27	0,02	0,6737	0,004789
nC_6		**0,0015**	1,0430	0,07	0,02	1,0459	0,008760
nC_7		0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_8		0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_9		0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_{10}		0,0000	0,0000	0,00	0,00	0,0000	0,000000
He		0,0000	0,0000	0,00	0,00	0,0000	0,000000
Ar		0,0000	0,0000	0,00	0,00	0,0000	0,000000
P_1	bara	**34,01**				0,0603	1,089926
T_1	°C	**20,00**				0,9502	-0,091585
		0					0
Rho, Sm3/h		**28,559174**				0,12%	

Tb, oC	**15,00**
Pb, bar	**1,0325**

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
	Equation of state	0,10	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,03	Normal	2,00	0,82971	0,02	1,94E-04
N	Nitrogen	0,36	Normal	2,00	0,00001	0,18	2,44E-12
CO_2	Carbon Dioxide	0,31	Normal	2,00	0,00019	0,15	8,54E-10
C_2	Ethane	0,43	Normal	2,00	0,07348	0,22	2,53E-04
C_3	Propane	0,31	Normal	2,00	0,05397	0,16	7,14E-05
iC_4	i-Butane	0,36	Normal	2,00	0,01627	0,18	8,46E-06
nC_4	n-Butane	0,34	Normal	2,00	0,01498	0,17	6,63E-06
iC_5	i-Pentane	0,58	Normal	2,00	0,00621	0,29	3,28E-06
nC_5	n-Pentane	0,67	Normal	2,00	0,00381	0,34	1,65E-06
nC_6	n-Hexane	1,05	Normal	2,00	0,00676	0,52	1,25E-05
u(CV)	Combined standard uncertainty		Normal			0,055	
U(CV)	Expanded uncertainty		Normal (k=2)	2,00		**0,11**	

CV Calculated in accordance with ISO6976-1995

Sensitivity coefficient calculation

	Value %	Composition Uncertainty %	Chromat Tolerance %	Repeatability %	Combined Uncertainty %	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,829712
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,000009
CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,000191
C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,073476
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,053968
H_2O	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2S	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
CO	0,0000	0,0000	0,00	0,00	0,0000	0,000000
O_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,016265
nC_4	**0,0050**	0,3228	0,10	[illegible]	0,3438	0,014980
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,006213
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,003810
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,006763
nC_7	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_8	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_9	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_{10}	0,0000	0,0000	0,00	0,00	0,0000	0,000000
He	0,0000	0,0000	0,00	0,00	0,0000	0,000000
Ar	0,0000	0,0000	0,00	0,00	0,0000	0,000000

CV MJ/m³ **40,6645618** **0,0470%**

Tb, °C	15	
Pb, Bar	1,01325	
Tz, °C	15	15
Thv, °C	15	0
	ISO6976-1995	**ISO6976-1983**
Hs.MJ/Sm³.10⁻⁵	40,66456181	42,84289135
R den	0,649235281	0,649198256
M mass. Kg/kmo	18,76090567	18,7604985
Std.Rho, kg/Sm³	0,795579211	0,79557667
Z	0,997313503	
Hs.kJ/kg	51,11315282	
Hs. kJ/mol	958,9290385	
Hi.MJ/Sm³.10⁻⁵	36,74175267	

PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
M	Ultrasonic calibration	0,298	Normal	2,00	1,00	0,15	0,022
Rho	Relation density	0,159	Normal	2,00	1,00	0,08	0,00[illegible]
Rhostd	Standard density	0,111	Normal	2,00	1,00	0,06	0,0[illegible]
Fc	Flow computer calculation	0,001	Normal	2,00	1,00	0,00	2,50E-07
$u_{(Qstd)}$	Combined standard uncertainty		Normal			0,178	
$U_{(Qstd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,36**	

PHU MY GDC - FQI 1420 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
qstd	Standard flowrate	0,356	Normal	2,00	1,00	0,18	0,032
CV	Calorific value	0,110	Normal	2,00	1,00	0,06	0,003
u(E)	Combined standard uncertainty		Normal			0,186	
U(E)	Expanded uncertainty		Normal (k=2)	2,00		**0,37**	

%Qmax	Flowrate	Utot	Error,%	Std J/R	E Std J/R
5,0	250,3	0,00	0,15	8818	3585578,0
10,1	503,5	0,00	0,17	17739	7211338,9
25,2	1262,4	0,00	0,17	44473	18084390,1
39,9	1996,4	0,00	0,15	70339	28599928,2
69,8	3488,2	0,00	0,14	122884	49970013,2
80,4	4021,7	0,00	0,15	141678	57612566,6

Pf, bar		34,01325	
Tf, °C		20	
Zf		0,91671	
Pb, bara		1,0325	
Tb, °C		15	
Zb		0,99731	
Hs MJ/Sm³ 10^{-6}		40,66456	

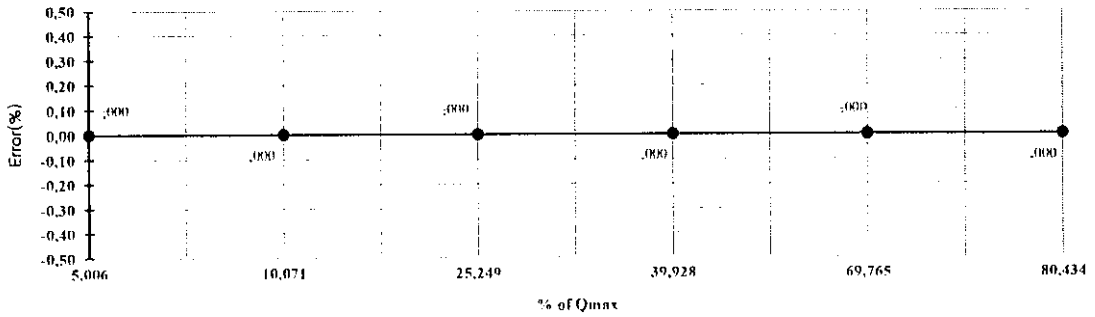
Standard Flowrate Expand Uncertainty

%Qmax	U_{metAct}	U_i	$(U_i \times C_i)^2$	U_{dev}	U_i	$(U_i \times C_i)^2$	$U_{thermal}$	U_i	$(Ui \times Ci)^2$	U_{fc}	U_i	$(U_i \times C_i)^2$	S^0,5	U
5,0	0,29	0,14	0,02	0,16	0,08	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,17	0,[illegible]
10,1	0,30	0,15	0,02	0,16	0,08	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,18	0,36
25,2	0,30	0,15	0,02	0,16	0,08	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,18	0,36
39,9	0,29	0,14	0,02	0,16	0,08	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,17	0,35
69,8	0,28	0,14	0,02	0,16	0,08	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,17	0,34
80,4	0,29	0,14	0,02	0,16	0,08	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,17	0,35
Distribution	Normal			Normal			Normal			Normal				Normal
Divisor	2			2			2			2				2
Ci	1			1			1			1				

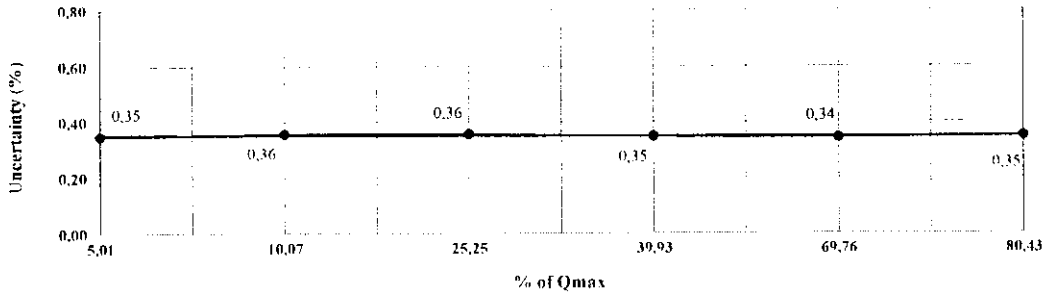
Energy Flowrate Expanded Uncertainty

%Qmax	$U_{EnergyStd}$	U_i	$(U_i \times C_i)^2$	Ucv	U_i	$(U_i \times C_i)^2$	S^0,5	U_{E_plate}
5,0	0,35	0,17	0,03	0,11	0,06	0,003	0,18	0,36
10,1	0,36	0,18	0,03	0,11	0,06	0,003	0,19	0,37
25,2	0,36	0,18	0,03	0,11	0,06	0,003	0,19	0,37
39,9	0,35	0,17	0,03	0,11	0,06	0,003	0,18	0,36
69,8	0,34	0,17	0,03	0,11	0,06	0,003	0,18	0,36
80,4	0,35	0,17	0,03	0,11	0,06	0,003	0,18	0,36
Distribution	Normal			Normal				Normal
Divisor	2			2				2
Ci	1			1				

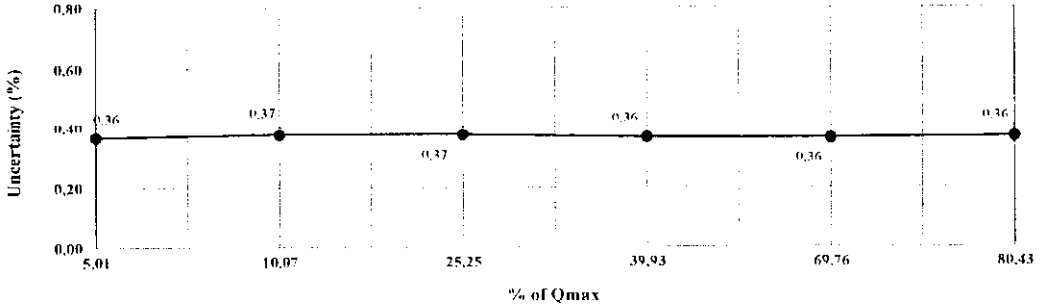
Error Curve



Standard Flowrate Expanded Uncertainty



Energy Flowrate Expanded Uncertainty



QUATEST 3®

TỔNG CỤC TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG
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TRUNG TÂM KỸ THUẬT TIÊU CHUẨN ĐO LƯỜNG CHẤT LƯỢNG 3
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UNCERTAINTY CALCULATION
STREAM FQI-1440

2020

PHU MY GDC - FQI 1440 - GAS METERING SYSTEM
Uncertainty Summary Report

	ISO 5168 - 2005
AGA8-94 (N - using AGA8-85)	Y
ISO 6976-95 (N - using ISO6976-83)	Y

Measurements

Line Pressure	34,01	bar a
Line Temperature	20,00	°C

Pressure	PT-1443
Static	
Manufacturer	Rosemount
Model No	3051STCG
Upper range limit	275,8 bar
Span	60,0 bar
SN	2732618

Composition

	% mole	Cal Gas Accuracy	Cal Gas Accuracy (% y)	GC Meas (% Mol)	% tolerance
Methane (C_1)	88,7970%	0,0210	0,0236	88,8070	0,0113
Nitrogen (N)	0,4849%	0,0017	0,3506	0,4846	0,0619
Carbon Dioxide (CO_2)	2,4555%	0,0071	0,2891	2,4576	0,0855
Ethane (C_2)	4,4840%	0,0170	0,3791	4,4759	0,1806
Propane (C_3)	2,3161%	0,0069	0,2979	2,3180	0,0820
Water (H_2O)	[illegible]	[illegible]	[illegible]	[illegible]	0,0000
Hydrogen Sulphide (H_2S)	[illegible]	[illegible]	[illegible]	[illegible]	0,0000
Hydrogen (H_2)	[illegible]	[illegible]	[illegible]	[illegible]	0,0000
Carbon Monoxide (CO)	[illegible]	[illegible]	[illegible]	[illegible]	0,0000
Oxygen (O_2)	[illegible]	[illegible]	[illegible]	[illegible]	0,0000
i-Butane (iC_4)	0,5400%	0,0018	0,3333	0,5406	0,1111
n-Butane (nC_4)	0,4956%	0,0016	0,3228	0,4961	0,1009
i-Pentane (iC_5)	0,1676%	0,00094	0,5608	0,1674	0,1372
n-Pentane (nC_5)	0,1025%	0,0006	0,5950	0,1028	0,2731
n-Hexane (nC_6)	0,1534%	0,0016	1,0430	0,1535	0,0652
n-Heptane (nC_7)	[illegible]		[illegible]	[illegible]	0,0000
n-Octane (nC_8)	[illegible]		[illegible]	[illegible]	0,0000
n-Nonane (nC_9)	[illegible]		[illegible]	[illegible]	0,0000
n-Decane (nC_{10})	[illegible]		[illegible]	[illegible]	0,0000
Helium (He)	[illegible]		[illegible]	[illegible]	0,0000
Argon (Ar)	[illegible]		[illegible]	[illegible]	0,0000
Sum	99,9967%				

Chromatograph repeatability 0,02 %

	Status
Reference Accuracy	0,025 %span
Transmitter calibration	0,063 % span
Ambient temperature effect	
%URL	0,009 % URL
%Span	0,025 % span
at	28 °C
with max. dev. temperature	10 °C
Transm. Cal. Tolerance	0,044 % span
Drift (stablity)	0,125 % URL
with stable duration	60 months
and calibration interval	12 months
ADC Tolerance	0,050 % span
Supply voltage effect	
Less than,% span	0,005 % span
Voltage variation	5,0 V

Equation of state

Compressibility factor, AGA8	0,1 %
Standard density, ISO6976	0,1 %
Calorific value, ISO6976	0,1 %

Uncertainty of correct factor

Uncertainty Limit 1 %

Temperature	Transmitter	RTD
Manufacturer	Rosemount	Rosemount
Model	3144P DIA	R68
Type	RTD	P1100
Span	80 °C	(-200 - 400)
SN	1067571	

	Trans
Digital Accuracy	0,1 °C
D/A Accuracy	0,02 % span
Transmitter Calibration Tolerance	0,05 °C
Ambient temperature effect	
with Max. temp deviation	10 °C
Digital accuracy (°C / °C)	0,0015 °C/°C
D/A effect (%span / °C)	0,01 % span /10 °C
Drift (stablity)	0,1 % reading
with stable duration	24 months
and calibration interval	12 months
Transmitter calibration	0,175 % span
Power supply effect	0,005 % span/volt
with voltage variation	5 v
ADC Tolerance	0,01 % span

FlowComputer	FQI-1440
Manufacturer	Daniel
Model	FloBoss S600
SN	NA
ADC resolution	32 bit
ADC accuracy	- %
Input resistor	- %
Calculation accuracy	0,001 %

USM/TB	FT-1440
Manufacturer	Daniel
Meter type	Ultrasonic
Model	3400
G-rating	
SN	02-050174
Meter dia.	10 inches
Maximum Flowrate	5000 m3/h
Maximum Pressure	80
Impulses / m^3 (HF)	3582,21
Impulses / m^3 (LF)	3582,21

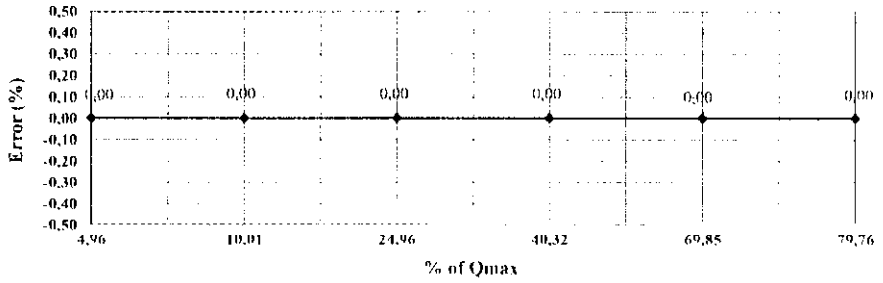
% Qmax	Flowrate (m^3/h)	U_{cal} (%)	U_{lin} (%)	U_{rel} (%)
5,0	247,79	0,14	0,000	0,30
10,0	500,27	0,13	0,000	0,30
25,0	1247,77	0,17	0,000	0,30
40,3	2015,98	0,18	0,000	0,30
69,8	3492,27	0,15	0,000	0,30
79,8	3987,93	0,17	0,000	0,30
		(Certificate)	(Certificate)	(AGA 9)

PHU MY GDC - FQI 1440 - GAS METERING SYSTEM

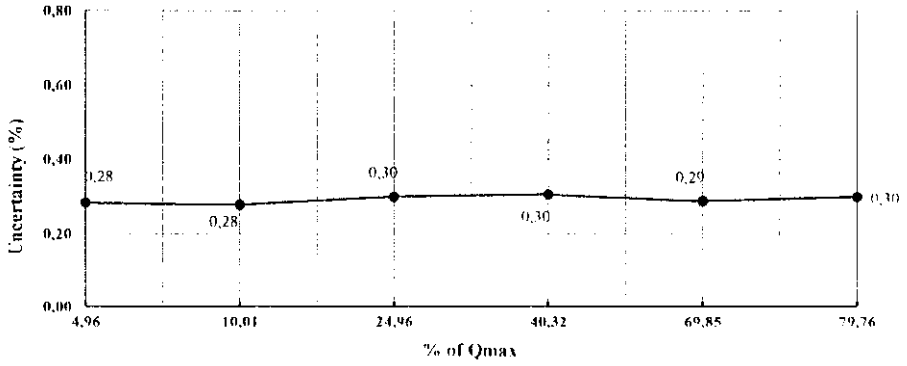
Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
U_{cal}	Ultrasonic calibration	0,180	Normal	2,00	1,00	0,09	8,10E-03
U_{tol}	Ultrasonic calibration tolerance	0,000	Rectangular	1,73	1,00	0,00	0,00E+00
U_{mi}	System effect	0,300	Triangular	2,45	1,00	0,12	1,50E-02
u(M)	Combined standard uncertainty		Normal			0,152	
U(M)	Expanded uncertainty		Normal (k=2)	2,00		0,30	

% Qmax	Q(m^3/h)	Deviation (%)	U_{cal}(%)	U_{tol}(%)	U_{mi}(%)	U_c(%)	$U_{exp(M)}$(%)
5,0	247,8	0,000	0,14	0,000	0,30	0,14	0,28
10,0	500,3	0,000	0,13	0,000	0,30	0,14	0,28
25,0	1247,8	0,000	0,17	0,000	0,30	0,15	0,30
40,3	2016,0	0,000	0,18	0,000	0,30	0,15	0,30
69,8	3492,3	0,000	0,15	0,000	0,30	0,14	0,29
79,8	3987,9	0,000	0,17	0,000	0,30	0,15	0,30

Error Curve



USM Calibration



PHU MY GDC - FQI 1440 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
Upac	Reference accuracy	0,025	Rectangular	1,73	1,00	0,014	2,09E-04
Upcal	Transmitter calibration	0,114	Normal	2,00	1,00	0,057	3,26E-03
Utemp	Ambient temperature effect	0,043	Rectangular	1,73	1,00	0,025	6,21E-04
Uadc	ADC tolerance (FlowComp. Input)	0,091	Rectangular	1,73	1,00	0,052	2,75E-03
Uptol	Transmitter calibration tolerance	0,080	Rectangular	1,73	1,00	0,046	2,11E-03
Upd	Drift	0,209	Rectangular	1,73	1,00	0,121	1,46E-02
Upp	Power supply effect	0,045	Rectangular	1,73	1,00	0,026	6,89E-04
u(p)	Combined standard uncertainty		Normal			0,156	
U(p)	Expanded uncertainty		Normal (k=2)	2,00		**0,31**	

URL	275,79	bar
Span	60	bar
Reading	33,00	bar
Turndown	4,6 :1	

Reference Accuracy

		0,025	% span
	=	0,015	bar
	=	0,045	% reading

Transmitter calibration

Cert. of calibration		0,063	% span
	=	0,038	bar
	=	0,114	% reading

Ambient temperature effect

Maximum temperature deviation			10	°C
			28	°C
0,009	% of URL	=	0,025	bar/28°C
0,025	% of span	=	0,015	bar/28°C
		sum =	0,040	bar/28°C
temperature effect		=	0,014	bar
		=	0,043	% reading

ADC test

Cert. of calibration		0,050	% span
	=	0,030	bar
	=	0,091	% reading

Transmitter Calibration Tolerance

Cert. of calibration		0,044	% span
	=	0,026	bar
	=	0,080	% reading

Drift

Stability Duration		60	months
Calibration interval		12	months
Stability	=	0,125	% URL/60 months
	=	0,025	% URL/12 months
	=	0,069	bar/12 months
	=	0,069	bar
	=	0,209	% reading

Power supply effect

Voltage variation		5	Volts
Stability	=	0,005	% span/volt
	=	0,003	bar/volt
	=	0,015	bar
	=	0,045	% reading

PHU MY GDC FQI 1440 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
U_{idac}	Digital accuracy	0,500	Rectangular	1,73	1,00	0,289	8,35E-02
U_{ida}	D/A Accuracy	0,080	Rectangular	1,73	1,00	0,046	2,14E-03
U_{itcal}	Transmitter calibration	0,700	Normal	2,00	1,00	0,350	1,23E-01
U_{itcmp}	Ambient temperature effect	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{itol}	Transmitter calibration tolerance	0,250	Rectangular	1,73	1,00	0,144	2,08E-02
U_{id}	Drift	0,050	Rectangular	1,73	1,00	0,029	8,33E-04
U_{ip}	Power supply effect	0,100	Rectangular	1,73	1,00	0,058	3,33E-03
U_{iadc}	ADC test	0,040	Rectangular	1,73	1,00	0,023	5,33E-04
$u_{(t)}$	Combined standard uncertainty		Normal			0,484	
$U_{(t)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,97**	

Model	3144P DIA		
Type	RTD		
Span	80	°C	
Reading	20	°C	
	= 295,2	K	

Digital Accuracy

From data sheet: 0,1 °C
= 0,50 % reading

D/A Accuracy

From data sheet: 0,02 % span
0,02 °C
= 0,08 % reading

Transmitter Calibration Tolerance

Cert. of calibration
= 0,05 °C
= 0,2500000 % reading

Ambient temperature effect

Maximum temperature deviation: 10 °C
D/A effect (0.001%span / °C): 0,010 °C
= 0,0500000 % reading
0,010000 °C
Ambient temperature effect: 0,0500000 % reading

Drift

Calibration interval: 12 months
Stability Duration: 24 months
Stability: greater of 0,1 % reading/24 months
or 0,05 % reading/12 months

Transmitter calibration

Cert. Of calibration: 0,18 % span
0,14 °C
= 0,70 % reading

Power supply effect

Voltage variation: 5 Volts

Stability = 0,005 % span/volt
= 0,004 °C/volt
= 0,02 °C
= 0,1000 % reading

ADC test

Cert. Of calibration: 0,010 % span
0,008 °C
= 0,040 % reading

PHU MY GDC - FQI 1440 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,76345	0,02	1,64E-04
N	Nitrogen	0,358	Normal	2,00	0,00725	0,18	1,69E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,05779	0,15	7,83E-05
C_2	Ethane	0,433	Normal	2,00	0,07230	0,22	2,45E-04
C_3	Propane	0,313	Normal	2,00	0,05476	0,16	7,36E-05
iC_4	i-Butane	0,358	Normal	2,00	0,01683	0,18	9,06E-06
nC_4	n-Butane	0,344	Normal	2,00	0,01545	0,17	7,05E-06
iC_5	i-Pentane	0,583	Normal	2,00	0,00649	0,29	3,58E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00397	0,34	1,79E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00709	0,52	1,38E-05
$u_{(Rhostd)}$	Combined standard uncertainty		Normal			0,056	
$U_{(Rhostd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,11**	

Standard density calculated in accordance with ISO6976-1995

Sensitivity coefficient calculation

	Value	Cal gas Uncertainty %	Calibration Tolerance %	Chromat Repeatability %	Combined Uncertainty %	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,763455
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,007249
CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,057793
C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,072299
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,054761
H_2O	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2S	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
CO	0,0000	0,0000	0,00	0,00	0,0000	0,000000
O_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,016830
nC_4	**0,0050**	0,3228	0,10	0,02	0,3438	0,015451
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,006486
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,003969
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,007093
nC_7	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_8	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_9	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_{10}	0,0000	0,0000	0,00	0,00	0,0000	0,000000
He	0,0000	0,0000	0,00	0,00	0,0000	0,000000
Ar	0,0000	0,0000	0,00	0,00	0,0000	0,000000

Std Rho kg/m³ **0,795579** 0,0489%

Tb, °C	15	
Pb, Bar	1,01325	
Tz, °C	15	15
Thv, °C	15	0
Hs.MJ/Sm³.10⁻⁵	40,6645618	42,842891
R den	0,64923528	0,6491983
M mass. Kg/kmol	18,7609057	18,760498
Std.Rho., kg/Sm³	0,79557921	0,7955767
Z	0,9973135	
Hs.kJ/kg	51,1131528	
Hs. kJ/mol	958,929039	
Hi.MJ/Sm³.10⁻⁵	36,7417527	

PHU MY GDC - FQI 1440 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Distribution	Divisor	c_i	u_i	$(c_i \times u_i)^2$
P	Absolute pressure	0,121	Normal	2,00	1,08993	0,06	4,32E-03
T	Line temperature	0,969	Normal	2,00	-0,09158	0,48	1,97E-03
Z	Equation of state	0,100	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,034	Normal	2,00	0,90386	0,02	2,30[illegible]
N	Nitrogen	0,358	Normal	2,00	0,00756	0,18	1,84E-06
CO_2	Carbon Dioxide	0,306	Normal	2,00	0,06374	0,15	9,52E-05
C_2	Ethane	0,433	Normal	2,00	0,08757	0,22	3,60E-04
C_3	Propane	0,313	Normal	2,00	0,06617	0,16	1,07E-04
iC_4	i-Butane	0,358	Normal	2,00	0,02032	0,18	1,32E-05
nC_4	n-Butane	0,344	Normal	2,00	0,01859	0,17	1,02E-05
iC_5	i-Pentane	0,583	Normal	2,00	0,00778	0,29	5,14E-06
nC_5	n-Pentane	0,674	Normal	2,00	0,00479	0,34	2,60E-06
nC_6	n-Hexane	1,046	Normal	2,00	0,00876	0,52	2,10E-05
u(Zf)	Combined standard uncertainty		Normal			0,098	
U(Zf)	Expanded uncertainty		Normal (k=2)	2,00		**0,20**	

Density calculated in accordance with AGA8 1992

Sensitivity coefficient calculation

		Component Value	Cal gas Uncertainty %	Calibration Tolerance %	Chromat Repeatability %	Combined Uncertainty %	c_i
C_1		**0,8880**	0,0236	0,01	0,02	0,0336	0,903861
N		**0,0048**	[illegible]	0,06	0,02	0,3584	0,007561
CO_2		**0,0246**	0,2891	0,09	0,02	0,3062	0,063737
C_2		**0,0448**	0,3791	0,18	0,02	0,4332	0,087573
C_3		**0,0232**	0,2979	0,08	0,02	0,3133	0,066172
H_2O		0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2S		0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2		0,0000	0,0000	0,00	0,00	0,0000	0,000000
CO		0,0000	0,0000	0,00	0,00	0,0000	0,000000
O_2		0,0000	0,0000	0,00	0,00	0,0000	0,000000
iC_4		**0,0054**	0,3333	0,11	0,02	0,3577	0,020323
nC_4		**0,0050**	0,3228	0,10	0,02	0,3438	0,018591
iC_5		**0,0017**	0,5608	0,14	0,02	0,5831	0,007778
nC_5		**0,0010**	0,5950	0,27	0,02	0,6737	0,004789
nC_6		**0,0015**	1,0430	0,07	0,02	1,0459	0,008760
nC_7		0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_8		0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_9		0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_{10}		0,0000	0,0000	0,00	0,00	0,0000	0,000000
He		0,0000	0,0000	0,00	0,00	0,0000	0,000000
Ar		0,0000	0,0000	0,00	0,00	0,0000	0,000000
P_1	bara	**34,01**				0,1206	1,089926
T_1	°C	**20,00**				0,9686	-0,091585
		0					0
Rho, Sm3/h		**28,559174**				0,17%	
Tb, oC		**15,00**					
Pb, bar		**1,0325**					

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
	Equation of state	0,10	Normal	2,00	1,00000	0,05	2,50E-03
C_1	Methane	0,03	Normal	2,00	0,82971	0,02	1,94E-04
N	Nitrogen	0,36	Normal	2,00	0,00001	0,18	2,44E-12
CO_2	Carbon Dioxide	0,31	Normal	2,00	0,00019	0,15	8,54E-10
C_2	Ethane	0,43	Normal	2,00	0,07348	0,22	2,53E-04
C_3	Propane	0,31	Normal	2,00	0,05397	0,16	7,14E-05
iC_4	i-Butane	0,36	Normal	2,00	0,01627	0,18	8,46E-06
nC_4	n-Butane	0,34	Normal	2,00	0,01498	0,17	6,63E-06
iC_5	i-Pentane	0,58	Normal	2,00	0,00621	0,29	3,28E-06
nC_5	n-Pentane	0,67	Normal	2,00	0,00381	0,34	1,65E-06
nC_6	n-Hexane	1,05	Normal	2,00	0,00676	0,52	1,25E-05
u(CV)	Combined standard uncertainty		Normal			0,055	
U(CV)	Expanded uncertainty		Normal (k=2)	2,00		**0,11**	

CV Calculated in accordance with ISO6976-1995

Sensitivity coefficient calculation

	Value %	Composition Uncertainty %	Chromat Tolerance %	Repeatability %	Combined Uncertainty %	c_i
C_1	**0,8880**	0,0236	0,01	0,02	0,0336	0,829712
N	**0,0048**	0,3506	0,06	0,02	0,3584	0,000009
CO_2	**0,0246**	0,2891	0,09	0,02	0,3062	0,000191
C_2	**0,0448**	0,3791	0,18	0,02	0,4332	0,073476
C_3	**0,0232**	0,2979	0,08	0,02	0,3133	0,053968
H_2O	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2S	0,0000	0,0000	0,00	0,00	0,0000	0,000000
H_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
CO	0,0000	0,0000	0,00	0,00	0,0000	0,000000
O_2	0,0000	0,0000	0,00	0,00	0,0000	0,000000
iC_4	**0,0054**	0,3333	0,11	0,02	0,3577	0,016265
nC_4	**0,0050**	0,3228	0,10	0,02	0,3438	0,014980
iC_5	**0,0017**	0,5608	0,14	0,02	0,5831	0,006213
nC_5	**0,0010**	0,5950	0,27	0,02	0,6737	0,003810
nC_6	**0,0015**	1,0430	0,07	0,02	1,0459	0,006763
nC_7	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_8	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_9	0,0000	0,0000	0,00	0,00	0,0000	0,000000
nC_{10}	0,0000	0,0000	0,00	0,00	0,0000	0,000000
He	0,0000	0,0000	0,00	0,00	0,0000	0,000000
Ar	0,0000	0,0000	0,00	0,00	0,0000	0,000000

CV	MJ/m^3	**40,6645618**	**0,0470%**

Tb, °C	15	
Pb, Bar	1,01325	
Tz, °C	15	15
Thv, °C	15	0
	ISO6976-1995	**ISO6976-1983**
Hs.MJ/Sm³.10⁻⁵	40,66456181	42,84289135
R den	0,649235281	0,649198256
M mass. Kg/kmo	18,76090567	18,7604985
Std.Rho, kg/Sm³	0,795579211	0,79557667
Z	0,997313503	
Hs.kJ/kg	51,11315282	
Hs. kJ/mol	958,9290385	
Hi.MJ/Sm³.10⁻⁵	36,74175267	

PHU MY GDC - FQI 1440 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
M	Ultrasonic calibration	0,304	Normal	2,00	1,00	0,15	0,023
Rho	Relation density	0,196	Normal	2,00	1,00	0,10	0,010
Rhostd	Standard density	0,111	Normal	2,00	1,00	0,06	0,0[illegible]
Fc	Flow computer calculation	0,001	Normal	2,00	1,00	0,00	2,50E-07
$u_{(Qstd)}$	Combined standard uncertainty		Normal			0,189	
$U_{(Qstd)}$	Expanded uncertainty		Normal (k=2)	2,00		**0,38**	

PHU MY GDC - FQI 1440 - GAS METERING SYSTEM

Symbol	Source of Uncertainty	Value %	Probability Distribution	Divisor / Multiplier	c_i	u_i	$(c_i \times u_i)^2$
qstd	Standard flowrate	0,379	Normal	2,00	1,00	0,19	0,036
CV	Calorific value	0,110	Normal	2,00	1,00	0,06	0,003
u(E)	Combined standard uncertainty		Normal			0,197	
U(E)	Expanded uncertainty		Normal (k=2)	2,00		**0,39**	

%Qmax	Flowrate	Utot	Error,%	Std FR	E Std Fr
5,0	247,8	0,00	0,11	8729	354968,0
10,0	500,3	0,00	0,13	17624	716653,5
25,0	1243,8	0,00	0,17	43952	1787374,8
40,3	2016,0	0,00	0,18	71019	2887962,9
69,8	3492,3	0,00	0,15	123026	5002800,7
79,8	3987,9	0,00	0,17	140487	5712851,2

Pf, bar		34,01325	
Tf, °C		20	
Zf		0,91671	
Pb, bara		1,0325	
Tb, °C		15	
Zb		0,99731	
Hs MJ/Sm³ 10^{-3}		40,66456	

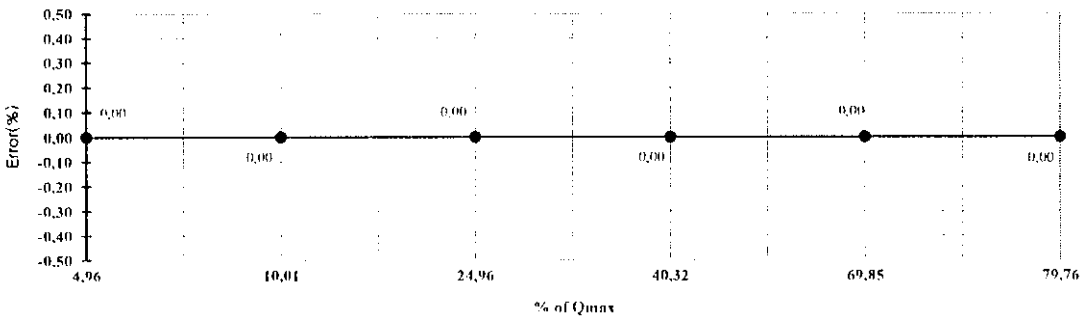
Standard Flowrate Expand Uncertainty

%Qmax	U_{meter}	U_1	$(U_1 \times C_i)^2$	U_{Sto}	U_2	$(U_2 \times C_i)^2$	U_{Dens}	U_3	$(U_3 \times C_i)^2$	U_{fc}	U_4	$(U_4 \times C_i)^2$	$S^2 0,5$	U_{Std}
5,0	0,28	0,14	0,02	0,20	0,10	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,18	0,36
10,0	0,28	0,14	0,02	0,20	0,10	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,18	0,36
25,0	0,30	0,15	0,02	0,20	0,10	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,19	0,37
40,3	0,30	0,15	0,02	0,20	0,10	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,19	0,38
69,8	0,29	0,14	0,02	0,20	0,10	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,18	0,37
79,8	0,30	0,15	0,02	0,20	0,10	0,01	0,11	0,06	0,003	0,001	0,001	3,E-07	0,19	0,37
Distribution	Normal			Normal			Normal			Normal				Normal
Divisor	2			2			2			2				2
Ci	1			1			1			1				

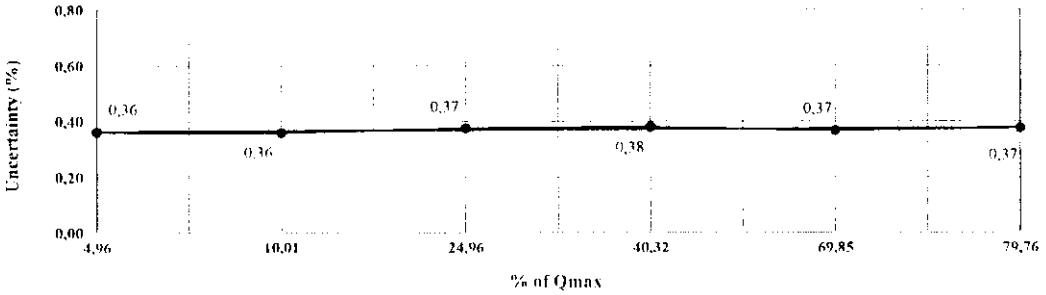
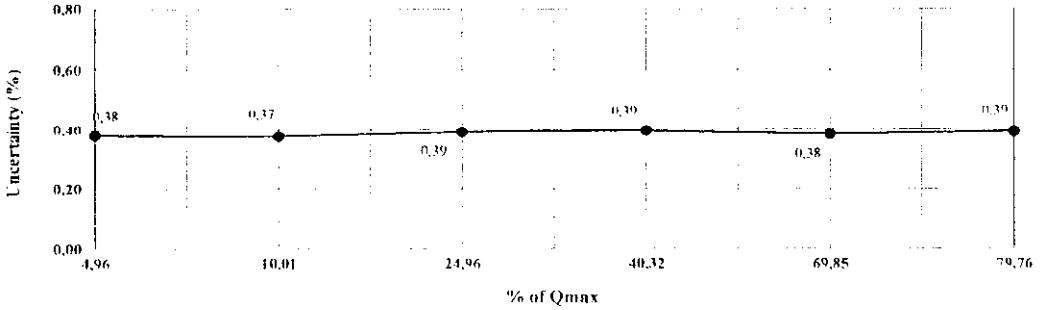
Energy Flowrate Expanded Uncertainty

%Qmax	$U_{Exp\,std}$	U_1	$(U_1 \times C_i)^2$	Ucv	U_2	$(U_2 \times C_i)^2$	$S^2 0,5$	$U_{E\,std}$
5,0	0,36	0,18	0,03	0,11	0,06	0,003	0,19	0,38
10,0	0,36	0,18	0,03	0,11	0,06	0,003	0,19	0,37
25,0	0,37	0,19	0,04	0,11	0,06	0,003	0,19	0,39
40,3	0,38	0,19	0,04	0,11	0,06	0,003	0,20	0,39
69,8	0,37	0,18	0,03	0,11	0,06	0,003	0,19	0,38
79,8	0,37	0,19	0,03	0,11	0,06	0,003	0,19	0,39
Distribution	Normal			Normal				Normal
Divisor	2			2				2
Ci	1			1				

Error Curve



Standard Flowrate Expanded Uncertainty



Energy Flowrate Expanded Uncertainty